

ADR V1 – Architecture Decision Record Repository

Section 1: Document Overview & Architectural Decision Governance

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Owner: Enterprise Architecture Function

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1. Purpose

The purpose of this Architecture Decision Record (ADR) document is to establish the governance framework, documentation standards, approval workflow, ownership model, and lifecycle management process for all architectural decisions within the Transparent Opportunity Decision Platform (TODP).

This document serves as the authoritative repository specification for recording, reviewing, approving, auditing, and maintaining architectural decisions that influence platform capabilities, scalability, security, governance, compliance, traceability, explainability, auditability, and replayability.

The ADR repository provides permanent institutional knowledge for architecture-related decisions and ensures decision transparency across engineering, product, governance, security, compliance, operations, and executive stakeholders.

2. Objectives

The objectives of ADR governance are:

OBJ-ADR-001

Provide a standardized mechanism for documenting architecture decisions.

OBJ-ADR-002

Maintain traceable relationships between business requirements, system requirements, architecture decisions, implementation artifacts, and operational controls.

OBJ-ADR-003

Ensure architecture decisions are reviewable, auditable, reproducible, and explainable.

OBJ-ADR-004

Reduce architectural drift across platform evolution.

OBJ-ADR-005

Support enterprise audit, compliance, and procurement reviews.

OBJ-ADR-006

Establish accountability for architecture ownership and decision approval.

OBJ-ADR-007

Enable future architectural replay and historical analysis.

OBJ-ADR-008

Support controlled technology adoption and retirement.

3. Scope

This ADR repository governs decisions affecting:

- Platform architecture
- Application architecture
- Service architecture
- Data architecture
- Security architecture
- AI architecture
- Governance architecture
- Audit architecture
- Integration architecture
- Deployment architecture
- Reliability architecture
- Compliance architecture
- Infrastructure architecture

The scope includes all architecture decisions impacting the following TODP modules:

- Opportunity
- Authority
- Evidence
- Retrieval
- Ranking
- Recommendation
- Governance
- Audit
- Replay

4. Architectural Decision Repository Purpose

The ADR repository acts as the official system of architectural record for TODP.

The repository shall:

- Capture architecture decisions
- Record alternatives considered
- Record decision rationale
- Document consequences
- Define implementation impacts

- Maintain decision status
- Preserve historical context
- Support future audits

No production-impacting architecture change may be implemented without a corresponding ADR.

5. Architectural Governance Principles

ADR governance shall follow the following principles:

PRN-ADR-001 Transparency

All significant architecture decisions must be documented.

PRN-ADR-002 Explainability

Decision rationale must be explicitly recorded.

PRN-ADR-003 Traceability

Each ADR must map to BRD, SRS, SAD, TDD, TEST, RTM, SEC, GOV, AUD, and REL artifacts where applicable.

PRN-ADR-004 Accountability

Each ADR must identify decision owners and approvers.

PRN-ADR-005 Auditability

Decision history must be immutable and reviewable.

PRN-ADR-006 Reproducibility

Future teams must understand why decisions were made.

PRN-ADR-007 Governance Alignment

Architecture decisions must align with approved governance policies.

6. ADR Classification Model

Architecture decisions shall be categorized using the following classifications:

ADR-ARCH

Core architecture decisions

ADR-DATA

Data architecture decisions

ADR-SEC
Security architecture decisions

ADR-AI
AI and model architecture decisions

ADR-OPS
Operational architecture decisions

ADR-INFRA
Infrastructure architecture decisions

ADR-COMP
Compliance-related architecture decisions

ADR-GOV
Governance architecture decisions

ADR-INTEG
Integration architecture decisions

ADR-DEPLOY
Deployment architecture decisions

7. ADR Lifecycle

Each ADR shall progress through the following states:

Draft

Initial proposal stage.

Under Review

Architecture review board evaluation.

Approved

Approved for implementation.

Implemented

Implemented within production systems.

Superseded

Replaced by a newer ADR.

Deprecated

No longer applicable.

Archived

Retained for historical and audit purposes.

Lifecycle transitions must be recorded and auditable.

8. Architecture Review Board Governance

The Architecture Review Board (ARB) shall be responsible for:

- Reviewing proposed ADRs
- Evaluating architecture risks
- Assessing technical tradeoffs
- Validating enterprise alignment
- Approving architecture changes
- Managing architectural consistency

The ARB shall maintain meeting records and decision logs.

9. Approval Requirements

The following approvals are mandatory before ADR approval:

Architecture Lead

Required

Engineering Lead

Required

Security Representative

Required for security-impacting ADRs

Governance Representative

Required for governance-impacting ADRs

Compliance Representative

Required for regulated decisions

Executive Sponsor

Required for strategic architecture decisions

10. Traceability Requirements

Every ADR must maintain traceability to:

BRD Requirements

Business objectives impacted.

SRS Requirements

Functional requirements impacted.

SAD Components

Architectural components affected.

TDD Components

Technical implementations affected.

TEST Specifications

Verification coverage.

RTM Entries

Requirement traceability references.

SEC Controls

Security control mappings.

GOV Policies

Governance policy mappings.

AUD Controls

Audit control mappings.

REL Requirements

Release dependencies.

11. Architectural Constraints

All ADRs must comply with the following platform constraints:

AC-001

Maintain explainability.

AC-002

Maintain traceability.

AC-003

Maintain auditability.

AC-004

Support replay capability.

AC-005

Preserve governance controls.

AC-006

Avoid uncontrolled data duplication.

AC-007

Preserve security boundaries.

AC-008

Maintain modular architecture.

AC-009

Support enterprise scalability.

AC-010

Support future extensibility.

12. Documentation Requirements

Every ADR must include:

- Decision Identifier
- Title
- Status
- Context
- Problem Statement
- Options Considered
- Decision Outcome
- Rationale
- Consequences
- Risks

- Dependencies
- Traceability References
- Approval Records
- Revision History

Incomplete ADRs cannot receive approval.

13. Metrics

The following governance metrics shall be maintained:

MET-ADR-001

Approved ADR count.

MET-ADR-002

Pending ADR count.

MET-ADR-003

Average review duration.

MET-ADR-004

ADR implementation completion rate.

MET-ADR-005

ADR compliance rate.

MET-ADR-006

Architecture exception count.

MET-ADR-007

Superseded ADR percentage.

MET-ADR-008

Traceability coverage percentage.

14. Audit Requirements

Architecture governance audits shall verify:

- ADR existence
- Approval evidence
- Traceability completeness

- Architecture consistency
- Compliance mappings
- Decision ownership
- Decision history retention
- Repository integrity

Audit evidence shall be retained according to enterprise retention policies.

15. Compliance Requirements

This ADR governance process shall support alignment with:

- SOC 2
- ISO 27001
- ISO 9001
- NIST Cybersecurity Framework
- Internal Governance Standards
- Enterprise Procurement Reviews
- Customer Security Assessments
- Vendor Due Diligence Reviews

All architecture decisions impacting regulated controls must maintain compliance traceability.

16. Section Summary

This section establishes the governance foundation for the TODP Architecture Decision Record repository. It defines the purpose, objectives, scope, lifecycle, ownership, approval processes, traceability requirements, architectural constraints, metrics, audit controls, and compliance obligations required to manage architecture decisions throughout the lifecycle of the Transparent Opportunity Decision Platform.

ADR V1 – Architecture Decision Record Repository

Section 2: ADR Repository Structure & Record Standards

Document ID: ADR-V1-S02

Document Title: ADR Repository Structure & Record Standards

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1. Purpose

The purpose of this section is to define the standard repository structure, record format, metadata requirements, storage controls, version management procedures, naming conventions, and documentation standards governing Architecture Decision Records (ADRs) within the Transparent Opportunity Decision Platform (TODP).

This section establishes a consistent and auditable architecture knowledge management framework to ensure that all architectural decisions are documented, discoverable, traceable, reviewable, and maintainable throughout the platform lifecycle.

2. Objectives

OBJ-ADR-REP-001

Establish a standardized repository structure for architecture decisions.

OBJ-ADR-REP-002

Ensure consistency across all ADR records.

OBJ-ADR-REP-003

Enable efficient architecture discovery and retrieval.

OBJ-ADR-REP-004

Support enterprise audit requirements.

OBJ-ADR-REP-005

Enable architecture traceability across documentation assets.

OBJ-ADR-REP-006

Preserve historical architecture knowledge.

OBJ-ADR-REP-007

Support future architectural replay and review activities.

OBJ-ADR-REP-008

Ensure repository integrity and governance compliance.

3. Scope

This section governs:

- ADR repository organization
- ADR record structures
- ADR metadata requirements

- Repository ownership
- Repository access controls
- ADR version management
- Repository retention requirements
- Traceability relationships
- Repository audit requirements

The scope applies to all architecture decisions impacting:

- Opportunity
- Authority
- Evidence
- Retrieval
- Ranking
- Recommendation
- Governance
- Audit
- Replay

4. Repository Architecture

The ADR repository shall function as the authoritative architecture decision repository for TODP.

The repository shall maintain:

- Active ADR records
- Historical ADR records
- Superseded ADR records
- Archived ADR records
- Decision dependency mappings
- Traceability mappings
- Approval records
- Audit evidence

The repository shall support controlled access and immutable decision history retention.

5. Repository Structure

The ADR repository shall be logically organized as follows:

/ADR

/Approved

/Under-Review

/Draft

/Superseded

/Deprecated

/Archived

/Decision-Templates

/Decision-Mappings

/Review-Records

/Audit-Evidence

Each repository segment shall maintain independent lifecycle controls and retention policies.

6. ADR Identification Standards

Every Architecture Decision Record shall receive a unique identifier.

Format:

ADR-XXXX

Examples:

ADR-0001

ADR-0002

ADR-0003

Identifier allocation shall be sequential and immutable.

Identifiers shall never be reused.

Deleted identifiers are prohibited.

7. ADR Metadata Standards

Every ADR shall contain mandatory metadata.

Required Fields:

ADR Identifier

Decision Title

Decision Category

Decision Status

Author

Owner

Review Authority

Approval Authority

Creation Date

Approval Date

Implementation Date

Supersession Status

Risk Classification

Affected Components

Affected Modules

Related Documents

Compliance Impact

Security Impact

Governance Impact

Audit Impact

Retention Classification

Metadata records shall be machine-readable wherever technically feasible.

8. ADR Naming Conventions

All ADR records shall follow standardized naming conventions.

Format:

ADR-XXXX-Decision-Title

Examples:

ADR-0007-Retrieval-Engine-Selection

ADR-0014-Evidence-Storage-Strategy

ADR-0022-Audit-Replay-Architecture

Naming conventions shall remain stable throughout the ADR lifecycle.

9. ADR Template Standards

All architecture decisions shall use an approved enterprise template.

Mandatory Template Sections:

Decision Identifier

Decision Title

Status

Context

Problem Statement

Business Drivers

Architecture Drivers

Options Considered

Decision Outcome

Decision Rationale

Tradeoff Analysis

Consequences

Risk Assessment

Dependencies

Implementation Requirements

Compliance Impact

Security Impact

Governance Impact

Approval Records

Revision History

Incomplete templates shall not progress to approval.

10. Decision Categorization Standards

Architecture decisions shall be classified into one primary category.

Categories:

Architecture

Data

Security

AI

Governance

Audit

Infrastructure

Operations

Integration

Deployment

A decision may contain secondary classifications where applicable.

11. Repository Ownership Model

The repository ownership structure shall be defined as follows:

Enterprise Architecture Function

Repository Owner

Architecture Review Board

Governance Authority

Engineering Leadership

Implementation Authority

Security Team

Security Validation Authority

Compliance Team

Regulatory Validation Authority

Product Governance Council

Final Governance Authority

Ownership assignments shall be documented and reviewed annually.

12. Version Management Standards

ADR version control shall preserve complete historical integrity.

Version States:

v1.0

Initial Approval

v1.1+

Minor Updates

v2.0+

Major Revisions

Version changes shall include:

- Revision description
- Author
- Reviewer
- Approval evidence
- Effective date

Historical versions shall remain accessible.

13. Traceability Record Standards

Each ADR shall maintain traceability relationships to:

BRD

Business Requirements

SRS

System Requirements

SAD

Solution Architecture

TDD

Technical Design

TEST

Testing Specifications

RTM

Requirement Traceability Matrix

SEC

Security Controls

GOV

Governance Controls

AUD

Audit Controls

REL

Release Records

Traceability gaps shall be classified as governance exceptions.

14. Repository Access Control Requirements

Repository access shall follow role-based access control.

Access Levels:

Read Only

Architecture Stakeholders

Contributor

Architecture Authors

Reviewer

ARB Members

Approver

Governance Authorities

Administrator

Repository Administrators

Access permissions shall be reviewed quarterly.

15. Repository Retention Requirements

ADR records shall be retained throughout the operational lifetime of TODP.

Minimum retention requirements:

Active ADRs

Permanent

Superseded ADRs

Permanent

Archived ADRs

Permanent

Approval Records

Permanent

Audit Evidence

Permanent

Deletion of ADR history is prohibited unless required by law.

16. Architectural Constraints

The repository shall satisfy the following constraints:

ARC-REP-001

Maintain immutable decision history.

ARC-REP-002

Support enterprise traceability.

ARC-REP-003

Support audit review.

ARC-REP-004

Support governance review.

ARC-REP-005

Support architecture replay.

ARC-REP-006

Maintain repository consistency.

ARC-REP-007

Prevent unauthorized modification.

ARC-REP-008

Preserve decision lineage.

17. Metrics

The following metrics shall be monitored:

MET-REP-001

Total ADR count.

MET-REP-002

Approved ADR count.

MET-REP-003

Draft ADR count.

MET-REP-004

Review cycle duration.

MET-REP-005

Traceability completeness.

MET-REP-006

Repository audit findings.

MET-REP-007

ADR compliance rate.

MET-REP-008

Repository integrity violations.

18. Audit Requirements

Repository audits shall verify:

- ADR completeness
- Metadata compliance
- Version history integrity
- Approval evidence existence
- Traceability coverage
- Repository access controls
- Retention compliance
- Governance compliance

Audit findings shall be documented and tracked to closure.

19. Compliance Requirements

The ADR repository shall support:

- SOC 2 evidence collection
- ISO 27001 documentation controls
- ISO 9001 change governance
- Enterprise procurement reviews
- Customer architecture assessments
- Internal audit programs
- Regulatory reviews where applicable

Compliance mappings shall be maintained for every approved ADR.

20. Section Summary

This section defines the ADR repository structure, metadata standards, identification rules, naming conventions, templates, version controls, ownership model, traceability requirements, access controls, retention policies, metrics, audit obligations, and compliance requirements necessary to manage architecture decisions consistently across the Transparent Opportunity Decision Platform. The repository serves as the authoritative architectural system of record and preserves complete decision lineage throughout the lifecycle of TODP.

ADR V1 – Architecture Decision Record Repository

Section 3: Architecture Principles & Decision Criteria

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1. Purpose

The purpose of this section is to establish the architectural principles, evaluation criteria, decision-making framework, tradeoff assessment methodology, and governance controls used to guide all architecture decisions within the Transparent Opportunity Decision Platform (TODP).

These principles ensure that architecture decisions remain aligned with platform objectives, business strategy, governance requirements, security obligations, compliance commitments, and long-term operational sustainability.

This section serves as the authoritative decision framework against which all Architecture Decision Records (ADRs) shall be evaluated and approved.

2. Objectives

OBJ-ARC-001

Establish a common architecture decision framework.

OBJ-ARC-002

Promote consistency across architecture decisions.

OBJ-ARC-003

Reduce subjective decision-making.

OBJ-ARC-004

Ensure architecture aligns with enterprise goals.

OBJ-ARC-005

Support explainability and transparency.

OBJ-ARC-006

Enable scalable platform evolution.

OBJ-ARC-007

Support governance, audit, and compliance requirements.

OBJ-ARC-008

Create repeatable decision evaluation processes.

3. Scope

This framework applies to all architecture decisions affecting:

- Application architecture
- Data architecture
- Security architecture
- Infrastructure architecture
- AI architecture
- Governance architecture
- Audit architecture
- Integration architecture
- Deployment architecture
- Operations architecture

The framework governs decisions impacting:

- Opportunity
- Authority
- Evidence
- Retrieval
- Ranking
- Recommendation
- Governance
- Audit
- Replay

4. Core Architecture Principles

The TODP architecture shall be governed by the following foundational principles.

PRN-001 Transparency

Architecture decisions must be understandable and reviewable.

PRN-002 Explainability

System behavior must be explainable to stakeholders.

PRN-003 Traceability

All decisions must be traceable to requirements and controls.

PRN-004 Auditability

Architecture must support evidence generation and audit review.

PRN-005 Replayability

Decision outcomes must be reproducible through replay mechanisms.

PRN-006 Governance First

Governance controls shall be embedded into architecture design.

PRN-007 Security by Design

Security requirements shall be incorporated during design activities.

PRN-008 Compliance by Design

Compliance obligations shall be considered during architecture selection.

PRN-009 Scalability

Architecture shall support future growth without major redesign.

PRN-010 Maintainability

Architecture shall prioritize long-term operational sustainability.

5. Enterprise Alignment Principles

Architecture decisions shall align with:

- Business objectives
- Product strategy
- Governance policies
- Security standards
- Compliance obligations
- Operational capabilities
- Customer requirements
- Enterprise risk tolerances

Architectural choices that conflict with approved enterprise objectives shall require executive review.

6. TODP-Specific Architectural Principles

The platform shall maintain the integrity of the core decision flow:

Opportunity

→ Authority

→ Evidence

→ Retrieval

→ Ranking

→ Recommendation

→ Governance Review

→ Audit

→ Replay

Architecture decisions shall not violate:

- Decision transparency
- Evidence traceability
- Governance enforcement
- Audit record generation
- Replay capability

Architectural changes impacting the core flow require formal ADR approval.

7. Modularity Principles

The architecture shall maintain strict module separation.

Required module boundaries:

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Replay

Cross-module coupling shall be minimized.

Direct dependencies shall be explicitly documented.

Hidden dependencies are prohibited.

8. Data Architecture Principles

Data-related decisions shall satisfy:

DAP-001

Single source of truth.

DAP-002

Controlled data ownership.

DAP-003

Data lineage visibility.

DAP-004

Data retention governance.

DAP-005

Data quality validation.

DAP-006

Data classification support.

DAP-007

Audit trail generation.

DAP-008

Replay support.

Data duplication shall require documented justification.

9. Security Architecture Principles

Security-related decisions shall follow:

SAP-001

Least privilege access.

SAP-002

Defense in depth.

SAP-003

Zero trust assumptions.

SAP-004

Encryption by default.

SAP-005

Identity-based authorization.

SAP-006

Security event logging.

SAP-007

Continuous monitoring.

SAP-008

Secure configuration management.

Security tradeoffs shall be formally documented.

10. Governance Architecture Principles

Governance decisions shall ensure:

GAP-001

Policy enforcement.

GAP-002

Approval workflows.

GAP-003

Exception management.

GAP-004

Decision accountability.

GAP-005

Evidence preservation.

GAP-006

Regulatory alignment.

GAP-007

Governance reporting.

GAP-008

Audit support.

Governance bypass mechanisms are prohibited unless formally approved.

11. Audit Architecture Principles

Audit-related decisions shall support:

AAP-001

Complete activity logging.

AAP-002

Immutable records.

AAP-003

Chain-of-custody preservation.

AAP-004

Historical reconstruction.

AAP-005

Replay capability.

AAP-006

Review traceability.

AAP-007

Evidence retention.

AAP-008

Investigation support.

Audit visibility shall not be compromised by architecture decisions.

12. Reliability Principles

Architecture decisions shall consider:

- Availability
- Durability
- Recoverability
- Fault isolation
- Operational resilience
- Service continuity
- Disaster recovery readiness

Single points of failure shall be identified and mitigated.

13. Scalability Principles

Architectural choices shall support:

- Increased users
- Increased opportunities
- Increased evidence volumes
- Increased retrieval loads
- Increased recommendation workloads
- Increased governance reviews
- Increased audit volumes

Scalability limitations shall be documented during decision evaluation.

14. Performance Principles

Architecture decisions shall evaluate:

- Response times
- Throughput
- Latency
- Resource consumption
- Cost efficiency
- User experience impact

Performance assumptions must be measurable and verifiable.

15. Integration Principles

Integration decisions shall support:

- Standardized interfaces
- Versioned contracts
- Controlled dependencies

- Observability
- Failure isolation
- Secure communication

Point-to-point integration proliferation shall be avoided.

16. Technology Selection Criteria

Technology evaluation shall consider:

TEC-001

Business alignment.

TEC-002

Architectural fit.

TEC-003

Operational maturity.

TEC-004

Security posture.

TEC-005

Compliance support.

TEC-006

Vendor viability.

TEC-007

Community adoption.

TEC-008

Scalability characteristics.

TEC-009

Maintainability.

TEC-010

Total cost of ownership.

Technology choices shall be documented within applicable ADRs.

17. Architecture Tradeoff Evaluation Framework

All significant architecture decisions shall evaluate:

Benefits

Expected value creation.

Risks

Potential negative impacts.

Alternatives

Competing approaches.

Costs

Implementation and operational costs.

Complexity

Engineering and operational complexity.

Governance Impact

Effect on governance processes.

Security Impact

Effect on security controls.

Compliance Impact

Effect on regulatory obligations.

Tradeoff analysis shall be mandatory for approval.

18. Decision Approval Criteria

An architecture decision may be approved when:

- Business alignment is demonstrated.
- Requirements traceability is complete.
- Security review is complete.
- Governance review is complete.
- Risks are documented.
- Alternatives were evaluated.

- Compliance implications are understood.
- Operational ownership is defined.

Approval criteria shall be verified by the Architecture Review Board.

19. Metrics

The following metrics shall be maintained:

MET-ARC-001

Architecture compliance rate.

MET-ARC-002

Approved decision percentage.

MET-ARC-003

Exception rate.

MET-ARC-004

Security review completion rate.

MET-ARC-005

Governance review completion rate.

MET-ARC-006

Traceability coverage.

MET-ARC-007

Architecture defect rate.

MET-ARC-008

Architecture rework rate.

20. Audit Requirements

Architecture audits shall verify:

- Principle adherence
- Decision rationale completeness
- Tradeoff documentation
- Approval evidence
- Governance alignment

- Security review completion
- Compliance review completion
- Traceability integrity

Audit findings shall be tracked through formal remediation processes.

21. Compliance Requirements

Architecture decision processes shall support:

- SOC 2
- ISO 27001
- ISO 9001
- NIST Cybersecurity Framework
- Enterprise Governance Standards
- Customer Security Assessments
- Procurement Reviews

Compliance obligations shall be incorporated into architecture evaluation activities.

22. Section Summary

This section defines the architecture principles, decision-making framework, evaluation criteria, technology selection standards, tradeoff analysis methodology, approval requirements, governance obligations, audit controls, and compliance expectations governing all architecture decisions within TODP. These principles provide the foundation for consistent, transparent, traceable, auditable, and enterprise-ready architectural decision-making across the platform lifecycle.

ADR V1 – Architecture Decision Record Repository

Section 4: Opportunity Module Architecture Decisions

Document ID: ADR-V1-S04

Document Title: Opportunity Module Architecture Decisions

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1. Purpose

The purpose of this section is to formally document the architectural decisions governing the Opportunity Module within the Transparent Opportunity Decision Platform (TODP).

The Opportunity Module serves as the entry point of the TODP decision lifecycle and is responsible for opportunity intake, opportunity normalization, opportunity enrichment, opportunity classification, opportunity lifecycle management, and opportunity traceability.

This section establishes the approved architectural patterns, design constraints, technology-neutral decisions, governance controls, audit requirements, and future evolution principles applicable to the Opportunity Module.

2. Objectives

OBJ-OPP-001

Establish a consistent architecture for opportunity intake and processing.

OBJ-OPP-002

Ensure opportunity traceability throughout the TODP lifecycle.

OBJ-OPP-003

Support structured and unstructured opportunity ingestion.

OBJ-OPP-004

Enable scalable opportunity management.

OBJ-OPP-005

Support governance and audit requirements.

OBJ-OPP-006

Preserve opportunity lineage.

OBJ-OPP-007

Enable replay of opportunity evaluations.

OBJ-OPP-008

Support enterprise-grade operational reliability.

3. Scope

The Opportunity Module architecture governs:

- Opportunity creation
- Opportunity ingestion
- Opportunity updates
- Opportunity lifecycle state management
- Opportunity metadata management
- Opportunity classification
- Opportunity tagging

- Opportunity ownership assignment
- Opportunity traceability management

The module acts as the first stage in the TODP workflow:

Opportunity
 → Authority
 → Evidence
 → Retrieval
 → Ranking
 → Recommendation
 → Governance
 → Audit
 → Replay

4. Architectural Decision ADR-0001

Decision Title

Opportunity as Independent Domain Module

Status

Approved

Decision

The Opportunity Module shall exist as an independently governed architectural domain with clearly defined ownership, interfaces, responsibilities, and lifecycle controls.

Rationale

Opportunity processing represents a foundational business capability and requires isolation from downstream evaluation logic.

Consequences

Positive

- Improved maintainability
- Independent scaling
- Clear ownership boundaries

Negative

- Additional interface management requirements

5. Architectural Decision ADR-0002

Decision Title

Opportunity-Centric Processing Model

Status

Approved

Decision

All downstream evaluations shall originate from a uniquely identified Opportunity entity.

Rationale

Maintains a single decision anchor throughout the platform.

Consequences

Positive

- End-to-end traceability
- Consistent audit lineage
- Simplified replay reconstruction

Negative

- Increased dependency on opportunity integrity

6. Architectural Decision ADR-0003

Decision Title

Globally Unique Opportunity Identifier

Status

Approved

Decision

Every opportunity shall receive a globally unique immutable Opportunity ID.

Requirements

- Unique across the platform
- Non-reusable
- Persistent for lifecycle duration
- Retained after closure

Rationale

Supports auditability, traceability, and replay capabilities.

Consequences

Positive

- Reliable record correlation
- Strong lineage management

Negative

- Identifier governance requirements

7. Architectural Decision ADR-0004

Decision Title

Opportunity Lifecycle State Model

Status

Approved

Decision

The Opportunity Module shall manage explicit lifecycle states.

Approved States

Draft

Submitted

Validated

Active

Under Review

Recommended

Rejected

Closed

Archived

Rationale

Provides consistent operational governance and audit visibility.

Consequences

Positive

- Lifecycle transparency
- State traceability

Negative

- Additional workflow governance complexity

8. Architectural Decision ADR-0005

Decision Title

Separation of Opportunity Data and Evaluation Data

Status

Approved

Decision

Opportunity information shall remain logically separate from Authority, Evidence, Ranking, and Recommendation outputs.

Rationale

Preserves architectural modularity.

Consequences

Positive

- Reduced coupling
- Clear ownership boundaries

Negative

- Additional integration requirements

9. Architectural Decision ADR-0006

Decision Title

Opportunity Metadata Framework

Status

Approved

Decision

The module shall support extensible metadata attributes without requiring core architectural redesign.

Required Metadata Categories

- Business metadata
- Classification metadata
- Ownership metadata
- Governance metadata
- Audit metadata

Rationale

Supports future extensibility and enterprise growth.

Consequences

Positive

- Flexible evolution
- Reduced redesign risk

Negative

- Metadata governance overhead

10. Architectural Decision ADR-0007

Decision Title

Structured and Unstructured Opportunity Support

Status

Approved

Decision

The Opportunity Module shall support ingestion of:

- Structured opportunities
- Semi-structured opportunities
- Unstructured opportunities

Examples

Structured

Application forms

Semi-Structured

Spreadsheets

Unstructured

Documents, emails, reports

Rationale

Supports diverse enterprise opportunity sources.

Consequences

Positive

- Broad adoption capability

Negative

- Increased normalization complexity

11. Architectural Decision ADR-0008

Decision Title

Opportunity Validation Layer

Status

Approved

Decision

All opportunities shall pass through a validation layer before becoming active.

Validation Categories

- Schema validation
- Business rule validation
- Governance validation
- Data quality validation

Rationale

Improves downstream processing quality.

Consequences

Positive

- Higher data quality
- Reduced operational defects

Negative

- Additional processing overhead

12. Architectural Decision ADR-0009

Decision Title

Opportunity Ownership Assignment

Status

Approved

Decision

Each opportunity shall have a designated accountable owner.

Ownership Requirements

- Owner identifier
- Ownership history
- Ownership transfer audit trail

Rationale

Supports accountability and governance.

Consequences

Positive

- Improved responsibility tracking

Negative

- Ownership management processes required

13. Architectural Decision ADR-0010

Decision Title

Opportunity Event Tracking

Status

Approved

Decision

All opportunity lifecycle events shall generate auditable records.

Event Examples

- Creation
- Update
- Submission
- Approval
- Rejection
- Closure

Rationale

Supports audit and replay requirements.

Consequences

Positive

- Complete activity history

Negative

- Increased storage requirements

14. Architectural Decision ADR-0011

Decision Title

Replay-Compatible Opportunity History

Status

Approved

Decision

Opportunity state changes shall be retained to support future replay operations.

Requirements

- Historical snapshots
- Change history
- State transitions
- Event timestamps

Rationale

Supports platform replayability objectives.

Consequences

Positive

- Accurate historical reconstruction

Negative

- Long-term retention obligations

15. Architectural Decision ADR-0012

Decision Title

Governance-Controlled Opportunity Modification

Status

Approved

Decision

Critical opportunity fields shall be governed by controlled modification rules.

Protected Categories

- Opportunity identifiers
- Classification values
- Ownership assignments
- Governance status indicators

Rationale

Protects integrity of decision processes.

Consequences

Positive

- Strong governance controls

Negative

- Reduced operational flexibility

16. Architectural Constraints

The Opportunity Module shall comply with:

OPP-CON-001

Maintain immutable opportunity identifiers.

OPP-CON-002

Preserve opportunity lineage.

OPP-CON-003

Support audit traceability.

OPP-CON-004

Support replay reconstruction.

OPP-CON-005

Maintain module independence.

OPP-CON-006

Support governance controls.

OPP-CON-007

Support scalable growth.

OPP-CON-008

Avoid hidden dependencies.

17. Metrics

The following metrics shall be monitored:

MET-OPP-001

Opportunity creation volume.

MET-OPP-002

Validation success rate.

MET-OPP-003

Lifecycle completion rate.

MET-OPP-004

Ownership assignment compliance.

MET-OPP-005

Metadata completeness.

MET-OPP-006

Opportunity processing latency.

MET-OPP-007

Replay reconstruction success rate.

MET-OPP-008

Audit traceability completeness.

18. Audit Requirements

Audits shall verify:

- Opportunity identifier uniqueness
- Lifecycle integrity
- Ownership records
- Validation compliance
- Metadata completeness
- Event logging completeness
- Replay support readiness
- Governance control enforcement

Audit evidence shall be retained according to enterprise retention policies.

19. Compliance Requirements

The Opportunity Module architecture shall support:

- SOC 2 traceability requirements
- ISO 27001 governance controls
- ISO 9001 process management controls
- Enterprise audit requirements
- Customer due diligence reviews
- Procurement architecture reviews

Compliance mappings shall be maintained through RTM and governance records.

20. Section Summary

This section defines the approved architecture decisions governing the Opportunity Module, including module ownership, lifecycle management, identifier strategy, validation controls,

metadata standards, event tracking, replay support, governance enforcement, audit obligations, and compliance requirements. These decisions establish Opportunity as the authoritative starting point of the TODP decision lifecycle while preserving transparency, traceability, auditability, governance, and replayability across the platform.

ADR V1 – Architecture Decision Record Repository

Section 5: Authority Module Architecture Decisions

Document ID: ADR-V1-S05

Document Title: Authority Module Architecture Decisions

Version: 1.0

Status: Approved

Owner: Enterprise Architecture Function

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to formally document the architectural decisions governing the Authority Module within the Transparent Opportunity Decision Platform (TODP).

The Authority Module is responsible for determining whether a user, role, group, system actor, or governance entity possesses the authority required to participate in opportunity evaluation, evidence submission, review activities, recommendation workflows, governance actions, audit operations, and replay processes.

This section establishes the approved architectural patterns, authorization principles, governance controls, audit requirements, trust boundaries, and operational constraints applicable to Authority management within TODP.

2. Objectives

OBJ-AUTH-001

Establish a centralized authority determination capability.

OBJ-AUTH-002

Provide consistent authorization decisions across all modules.

OBJ-AUTH-003

Support enterprise governance requirements.

OBJ-AUTH-004

Maintain decision accountability.

OBJ-AUTH-005

Enable auditable authorization outcomes.

OBJ-AUTH-006

Support scalable access management.

OBJ-AUTH-007

Preserve traceability of authority decisions.

OBJ-AUTH-008

Support replay of authorization evaluations.

3. Scope

The Authority Module architecture governs:

- Identity-to-authority mapping
- Role evaluation
- Permission evaluation
- Authority validation
- Delegation management
- Approval authority determination
- Governance authority determination
- Access decision traceability
- Authorization audit records

The Authority Module operates immediately after Opportunity intake within the TODP workflow:

Opportunity

→ Authority

→ Evidence

→ Retrieval

→ Ranking

→ Recommendation

→ Governance

→ Audit

→ Replay

4. Architectural Decision ADR-0013

Decision Title

Authority as Independent Decision Domain

Status

Approved

Decision

Authority evaluation shall exist as a dedicated architectural domain independent of Opportunity, Evidence, Ranking, Recommendation, Governance, Audit, and Replay processing.

Rationale

Authorization logic changes independently from business opportunity logic and requires separate governance.

Consequences

Positive

- Reduced coupling
- Clear ownership
- Independent evolution

Negative

- Additional integration interfaces

5. Architectural Decision ADR-0014

Decision Title

Centralized Authorization Evaluation Model

Status

Approved

Decision

Authorization decisions shall be evaluated through a centralized Authority service rather than distributed authorization logic.

Rationale

Ensures consistent authorization behavior across the platform.

Consequences

Positive

- Consistency
- Simplified governance
- Reduced policy duplication

Negative

- Additional dependency management

6. Architectural Decision ADR-0015

Decision Title

Authority Determination Before Evidence Processing

Status

Approved

Decision

Authority validation shall occur before evidence submission, evidence retrieval, recommendation review, governance approval, and replay execution.

Rationale

Prevents unauthorized influence on decision outcomes.

Consequences

Positive

- Strong governance enforcement
- Reduced security risk

Negative

- Additional validation operations

7. Architectural Decision ADR-0016

Decision Title

Role-Based Authority Foundation

Status

Approved

Decision

The Authority Module shall implement a role-based authority model as the foundational authorization mechanism.

Core Concepts

- Role
- Permission
- Authority Scope
- Delegation
- Governance Privilege

Rationale

Provides predictable and manageable authorization structures.

Consequences

Positive

- Enterprise familiarity
- Simplified administration

Negative

- Role maintenance overhead

8. Architectural Decision ADR-0017

Decision Title

Attribute-Aware Authorization Extension

Status

Approved

Decision

The architecture shall support future attribute-based authorization extensions without redesigning the core Authority model.

Example Attributes

- Department
- Region
- Business Unit
- Classification Level
- Governance Scope

Rationale

Supports future enterprise scaling requirements.

Consequences

Positive

- Extensibility
- Future readiness

Negative

- Additional evaluation complexity

9. Architectural Decision ADR-0018

Decision Title

Authority Scope Segmentation

Status

Approved

Decision

Authority shall be segmented according to business responsibility domains.

Approved Authority Scopes

Opportunity Authority

Evidence Authority

Retrieval Authority

Ranking Authority

Recommendation Authority

Governance Authority

Audit Authority

Replay Authority

Rationale

Maintains separation of duties.

Consequences

Positive

- Reduced governance risk
- Better accountability

Negative

- More granular administration

10. Architectural Decision ADR-0019

Decision Title

Separation of Authentication and Authorization

Status

Approved

Decision

Authentication and authorization shall remain architecturally distinct functions.

Authentication

Identity verification.

Authorization

Authority determination.

Rationale

Supports security best practices and architectural clarity.

Consequences

Positive

- Improved maintainability
- Better security posture

Negative

- Additional integration requirements

11. Architectural Decision ADR-0020

Decision Title

Delegated Authority Model

Status

Approved

Decision

Authority assignments may be delegated through governed delegation mechanisms.

Delegation Requirements

- Explicit authorization
- Time limitation
- Audit records
- Approval tracking

Rationale

Supports operational continuity.

Consequences

Positive

- Operational flexibility

Negative

- Governance oversight requirements

12. Architectural Decision ADR-0021

Decision Title

Authority Decision Traceability

Status

Approved

Decision

Every authorization decision shall generate traceable records.

Required Elements

- Requestor
- Authority source
- Evaluated permissions

- Evaluation result
- Timestamp
- Decision rationale

Rationale

Supports auditability and governance review.

Consequences

Positive

- Full authorization visibility

Negative

- Additional storage requirements

13. Architectural Decision ADR-0022

Decision Title

Replay-Compatible Authority Evaluation

Status

Approved

Decision

Authorization decisions shall be reproducible during replay operations.

Requirements

- Historical role mappings
- Historical permissions
- Delegation history
- Evaluation timestamps

Rationale

Supports accurate reconstruction of historical decisions.

Consequences

Positive

- Replay integrity

Negative

- Retention overhead

14. Architectural Decision ADR-0023

Decision Title

Governance-Protected Privilege Management

Status

Approved

Decision

Privilege assignment and privilege escalation shall require governance controls.

Governed Activities

- Role assignment
- Privilege elevation
- Delegation creation
- Authority scope changes

Rationale

Protects against unauthorized privilege expansion.

Consequences

Positive

- Strong governance
- Reduced insider risk

Negative

- Additional approval workflows

15. Architectural Decision ADR-0024

Decision Title

Authority Policy Repository

Status

Approved

Decision

Authority evaluation rules shall be maintained in a centralized policy repository.

Repository Requirements

- Version control
- Audit history
- Governance approvals
- Change traceability

Rationale

Supports policy consistency.

Consequences

Positive

- Simplified governance

Negative

- Repository administration effort

16. Architectural Decision ADR-0025

Decision Title

Separation of Duties Enforcement

Status

Approved

Decision

The Authority Module shall enforce separation-of-duties controls where governance policies require independent review.

Examples

- Evidence submitter cannot approve governance outcome.
- Recommendation creator cannot perform final approval.
- Audit reviewer cannot modify audit evidence.

Rationale

Reduces conflict-of-interest risk.

Consequences

Positive

- Improved governance integrity

Negative

- Additional authorization complexity

17. Architectural Constraints

The Authority Module shall comply with:

AUTH-CON-001

Centralized authorization decisions.

AUTH-CON-002

Traceable authority outcomes.

AUTH-CON-003

Replay-compatible evaluations.

AUTH-CON-004

Governance-controlled privilege changes.

AUTH-CON-005

Support separation of duties.

AUTH-CON-006

Maintain module independence.

AUTH-CON-007

Prevent unauthorized escalation.

AUTH-CON-008

Preserve audit visibility.

18. Metrics

The following metrics shall be monitored:

MET-AUTH-001

Authorization request volume.

MET-AUTH-002

Authorization latency.

MET-AUTH-003

Denied authorization rate.

MET-AUTH-004

Privilege escalation requests.

MET-AUTH-005

Delegation volume.

MET-AUTH-006

Authority traceability coverage.

MET-AUTH-007

Replay authorization accuracy.

MET-AUTH-008

Governance policy compliance rate.

19. Audit Requirements

Audits shall verify:

- Role assignment integrity
- Permission assignment controls
- Delegation governance compliance
- Separation-of-duties enforcement
- Authorization traceability
- Authority repository integrity
- Privilege management controls
- Replay readiness

Audit evidence shall be retained according to enterprise retention requirements.

20. Compliance Requirements

The Authority Module architecture shall support:

- SOC 2 logical access controls
- ISO 27001 access management requirements
- ISO 27001 segregation-of-duty controls
- Internal governance standards
- Enterprise audit requirements
- Customer security assessments
- Regulatory review activities where applicable

Compliance mappings shall be maintained through RTM, SEC, GOV, and AUD artifacts.

21. Section Summary

This section defines the approved architecture decisions governing the Authority Module, including centralized authorization evaluation, role-based authority management, delegated authority controls, separation-of-duties enforcement, traceability requirements, governance-protected privilege management, replay support, audit controls, and compliance obligations. These decisions establish Authority as the trusted authorization layer of TODP and ensure that participation in all decision lifecycle activities remains governed, auditable, traceable, and reproducible.

ADR V1 – Architecture Decision Record Repository

Section 6: Evidence Module Architecture Decisions

Document ID: ADR-V1-S06

Document Title: Evidence Module Architecture Decisions

Version: 1.0

Status: Approved

Owner: Enterprise Architecture Function

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to formally document the architectural decisions governing the Evidence Module within the Transparent Opportunity Decision Platform (TODP).

The Evidence Module serves as the authoritative evidence management capability of TODP and is responsible for evidence ingestion, evidence validation, evidence classification, evidence lineage tracking, evidence retention, evidence governance, evidence retrieval support, and evidence auditability.

The module provides the factual foundation upon which retrieval, ranking, recommendation, governance, audit, and replay activities are performed.

2. Objectives

OBJ-EVD-001

Establish a trusted evidence repository.

OBJ-EVD-002

Maintain evidence integrity throughout its lifecycle.

OBJ-EVD-003

Support structured and unstructured evidence sources.

OBJ-EVD-004

Provide end-to-end evidence traceability.

OBJ-EVD-005

Enable governance-controlled evidence usage.

OBJ-EVD-006

Support audit and compliance requirements.

OBJ-EVD-007

Enable replay of historical evidence states.

OBJ-EVD-008

Support scalable evidence operations.

3. Scope

The Evidence Module architecture governs:

- Evidence ingestion
- Evidence registration
- Evidence classification
- Evidence storage
- Evidence validation
- Evidence enrichment
- Evidence lineage tracking
- Evidence retention
- Evidence governance
- Evidence lifecycle management

The module operates after Authority validation within the TODP workflow:

Opportunity

→ Authority

→ Evidence

- Retrieval
- Ranking
- Recommendation
- Governance
- Audit
- Replay

4. Architectural Decision ADR-0026

Decision Title

Evidence as Independent Domain Module

Status

Approved

Decision

The Evidence Module shall operate as an independent architectural domain with dedicated ownership, governance controls, lifecycle management, and audit requirements.

Rationale

Evidence represents a core enterprise asset and requires governance independent of downstream processing.

Consequences

Positive

- Clear ownership
- Improved governance
- Better maintainability

Negative

- Additional integration requirements

5. Architectural Decision ADR-0027

Decision Title

Evidence as Authoritative Source Layer

Status

Approved

Decision

Evidence shall be treated as the authoritative factual foundation supporting retrieval, ranking, recommendation, governance, audit, and replay functions.

Rationale

Ensures that all downstream decisions are grounded in verifiable information.

Consequences

Positive

- Increased decision quality
- Improved explainability

Negative

- Greater emphasis on evidence quality controls

6. Architectural Decision ADR-0028

Decision Title

Immutable Evidence Identifier Strategy

Status

Approved

Decision

Every evidence object shall receive a globally unique immutable Evidence Identifier.

Requirements

- Globally unique
- Non-reusable
- Persistent
- Auditable

Rationale

Supports lineage tracking and audit reconstruction.

Consequences

Positive

- Strong traceability
- Simplified correlation

Negative

- Identifier governance requirements

7. Architectural Decision ADR-0029

Decision Title

Structured and Unstructured Evidence Support

Status

Approved

Decision

The architecture shall support:

Structured Evidence

Database records

Semi-Structured Evidence

Spreadsheets, forms, XML, JSON

Unstructured Evidence

Documents, emails, reports, presentations, media assets

Rationale

Enterprise evidence originates from diverse sources.

Consequences

Positive

- Broad evidence coverage

Negative

- Increased ingestion complexity

8. Architectural Decision ADR-0030

Decision Title

Evidence Classification Framework

Status

Approved

Decision

All evidence shall be classified according to governance-defined categories.

Minimum Classification Dimensions

- Evidence Type
- Source Type
- Trust Level
- Sensitivity Level
- Business Domain
- Retention Category

Rationale

Supports governance and retrieval effectiveness.

Consequences

Positive

- Better discoverability
- Improved governance

Negative

- Additional metadata requirements

9. Architectural Decision ADR-0031

Decision Title

Evidence Validation Layer

Status

Approved

Decision

Evidence shall undergo validation before becoming eligible for retrieval and ranking activities.

Validation Categories

- Format validation
- Integrity validation
- Metadata validation
- Governance validation
- Source validation

Rationale

Improves reliability of downstream decisions.

Consequences

Positive

- Higher evidence quality

Negative

- Increased processing overhead

10. Architectural Decision ADR-0032

Decision Title

Evidence Lineage Preservation

Status

Approved

Decision

The Evidence Module shall maintain complete lineage information throughout the evidence lifecycle.

Lineage Elements

- Source
- Ingestion event
- Transformation history
- Classification history
- Access history
- Retention history

Rationale

Supports explainability, governance, and audits.

Consequences

Positive

- Complete evidence visibility

Negative

- Additional storage requirements

11. Architectural Decision ADR-0033

Decision Title

Evidence Version Management

Status

Approved

Decision

Evidence modifications shall be managed through version-controlled mechanisms.

Requirements

- Historical versions retained
- Change history recorded
- Version lineage maintained
- Prior states preserved

Rationale

Supports auditability and replayability.

Consequences

Positive

- Historical transparency

Negative

- Increased retention costs

12. Architectural Decision ADR-0034

Decision Title

Governance-Controlled Evidence Lifecycle

Status

Approved

Decision

Evidence shall be governed through explicit lifecycle states.

Approved States

Registered

Validated

Active

Restricted

Retired

Archived

Disposed

Rationale

Provides governance visibility and lifecycle consistency.

Consequences

Positive

- Strong lifecycle governance

Negative

- Additional operational controls

13. Architectural Decision ADR-0035

Decision Title

Evidence Ownership Model

Status

Approved

Decision

Each evidence object shall maintain accountable ownership.

Ownership Attributes

- Owner
- Steward
- Source System
- Governance Contact

Rationale

Supports accountability and issue resolution.

Consequences

Positive

- Improved governance

Negative

- Ownership maintenance effort

14. Architectural Decision ADR-0036

Decision Title

Evidence Access Governance

Status

Approved

Decision

Evidence access shall be controlled through Authority Module decisions and governance policies.

Access Controls

- Read access
- Usage authorization
- Restricted access
- Administrative access

Rationale

Protects evidence integrity and confidentiality.

Consequences

Positive

- Reduced access risk

Negative

- Additional authorization evaluations

15. Architectural Decision ADR-0037

Decision Title

Replay-Compatible Evidence Preservation

Status

Approved

Decision

Historical evidence states shall be retained to support replay operations.

Requirements

- Historical versions
- Historical classifications
- Historical metadata
- Historical lineage

Rationale

Supports accurate reconstruction of past decisions.

Consequences

Positive

- Replay integrity

Negative

- Long-term retention obligations

16. Architectural Decision ADR-0038

Decision Title

Evidence Usage Traceability

Status

Approved

Decision

All evidence usage shall generate auditable records.

Tracked Activities

- Retrieval
- Review
- Ranking input
- Recommendation input
- Governance review
- Replay usage

Rationale

Provides end-to-end evidence accountability.

Consequences

Positive

- Complete usage visibility

Negative

- Additional event storage

17. Architectural Decision ADR-0039

Decision Title

Evidence Trust Model

Status

Approved

Decision

Evidence shall maintain trust attributes that can be evaluated by downstream modules.

Trust Dimensions

- Source reliability
- Validation status
- Governance status
- Freshness status
- Completeness status

Rationale

Supports explainable ranking and recommendations.

Consequences

Positive

- Improved decision quality

Negative

- Additional metadata management

18. Architectural Constraints

The Evidence Module shall comply with:

EVD-CON-001

Maintain immutable evidence identifiers.

EVD-CON-002

Preserve evidence lineage.

EVD-CON-003

Support governance-controlled access.

EVD-CON-004

Support replay reconstruction.

EVD-CON-005

Maintain evidence integrity.

EVD-CON-006

Support audit traceability.

EVD-CON-007

Maintain module independence.

EVD-CON-008

Preserve historical evidence states.

19. Metrics

The following metrics shall be monitored:

MET-EVD-001

Evidence ingestion volume.

MET-EVD-002

Validation success rate.

MET-EVD-003

Classification completeness.

MET-EVD-004

Evidence access volume.

MET-EVD-005

Lineage completeness.

MET-EVD-006

Evidence freshness compliance.

MET-EVD-007

Replay evidence reconstruction accuracy.

MET-EVD-008

Evidence governance compliance rate.

20. Audit Requirements

Audits shall verify:

- Evidence identifier uniqueness
- Evidence lineage completeness
- Validation compliance
- Classification compliance
- Access governance controls
- Version retention compliance
- Usage traceability
- Replay readiness

Audit evidence shall be retained according to enterprise retention requirements.

21. Compliance Requirements

The Evidence Module architecture shall support:

- SOC 2 evidence management controls
- ISO 27001 information governance controls
- ISO 27001 asset management requirements
- Enterprise audit requirements
- Customer security assessments
- Regulatory review obligations where applicable
- Internal governance policies

Compliance mappings shall be maintained through RTM, SEC, GOV, AUD, and COMP artifacts.

22. Section Summary

This section defines the approved architecture decisions governing the Evidence Module, including evidence ownership, classification, validation, lineage preservation, version management, governance-controlled access, trust modeling, usage traceability, replay support, audit controls, and compliance obligations. These decisions establish Evidence as the authoritative factual foundation of TODP and ensure that all downstream decision-making remains transparent, traceable, auditable, governed, and reproducible.

ADR V1 – Architecture Decision Record Repository

Section 7: Retrieval Module Architecture Decisions

Document ID: ADR-V1-S07

Document Title: Retrieval Module Architecture Decisions

Version: 1.0

Status: Approved

Owner: Enterprise Architecture Function

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to formally document the architectural decisions governing the Retrieval Module within the Transparent Opportunity Decision Platform (TODP).

The Retrieval Module is responsible for identifying, locating, filtering, selecting, and returning evidence relevant to an opportunity under evaluation. The module acts as the controlled bridge between Evidence and downstream analytical processes, ensuring that only appropriate, traceable, authorized, and contextually relevant evidence is made available for ranking and recommendation activities.

This section defines the approved architecture decisions, retrieval principles, governance controls, audit requirements, and operational constraints governing retrieval operations.

2. Objectives

OBJ-RET-001

Provide accurate evidence retrieval.

OBJ-RET-002

Support explainable retrieval outcomes.

OBJ-RET-003

Maintain evidence traceability.

OBJ-RET-004

Enforce authorization and governance controls.

OBJ-RET-005

Support scalable retrieval operations.

OBJ-RET-006

Enable auditability of retrieval decisions.

OBJ-RET-007

Support replay reconstruction.

OBJ-RET-008

Preserve retrieval transparency.

3. Scope

The Retrieval Module architecture governs:

- Evidence discovery
- Retrieval request processing
- Retrieval filtering
- Relevance determination
- Retrieval authorization validation
- Retrieval result generation
- Retrieval traceability
- Retrieval logging
- Retrieval replay support

The module operates within the TODP workflow as follows:

Opportunity
→ Authority

- Evidence
- Retrieval
- Ranking
- Recommendation
- Governance
- Audit
- Replay

4. Architectural Decision ADR-0040

Decision Title

Retrieval as Independent Domain Module

Status

Approved

Decision

The Retrieval Module shall operate as an independent architectural domain with dedicated ownership, governance controls, operational metrics, and lifecycle management.

Rationale

Retrieval capabilities evolve independently from evidence storage and ranking methodologies.

Consequences

Positive

- Modular evolution
- Independent scaling
- Reduced coupling

Negative

- Additional interface management

5. Architectural Decision ADR-0041

Decision Title

Retrieval as Evidence Access Layer

Status

Approved

Decision

The Retrieval Module shall function as the controlled evidence access layer for downstream analytical modules.

Rationale

Provides a governed mechanism for accessing evidence without exposing evidence storage internals.

Consequences

Positive

- Better abstraction
- Improved governance

Negative

- Additional architectural layer

6. Architectural Decision ADR-0042

Decision Title

Opportunity-Aware Retrieval Context

Status

Approved

Decision

All retrieval operations shall be performed within the context of a specific Opportunity.

Required Inputs

- Opportunity ID
- Authority context
- Retrieval request context

Rationale

Ensures contextual relevance.

Consequences

Positive

- Improved retrieval precision
- Better traceability

Negative

- Increased contextual dependency

7. Architectural Decision ADR-0043

Decision Title

Authority-Validated Retrieval

Status

Approved

Decision

Retrieval operations shall only execute after successful Authority validation.

Validation Requirements

- Requestor authorization
- Evidence access permissions
- Governance restrictions

Rationale

Protects evidence access boundaries.

Consequences

Positive

- Strong access governance

Negative

- Additional validation processing

8. Architectural Decision ADR-0044

Decision Title

Multi-Strategy Retrieval Architecture

Status

Approved

Decision

The architecture shall support multiple retrieval strategies.

Approved Strategy Categories

- Metadata retrieval
- Keyword retrieval
- Semantic retrieval
- Hybrid retrieval
- Filter-based retrieval

Rationale

Different evidence types require different retrieval approaches.

Consequences

Positive

- Increased retrieval flexibility

Negative

- Additional operational complexity

9. Architectural Decision ADR-0045

Decision Title

Retrieval Strategy Abstraction Layer

Status

Approved

Decision

Retrieval logic shall be abstracted from retrieval strategy implementations.

Rationale

Supports future retrieval innovation without redesigning consumers.

Consequences

Positive

- Improved extensibility

Negative

- Additional abstraction overhead

10. Architectural Decision ADR-0046

Decision Title

Evidence Metadata-Driven Retrieval

Status

Approved

Decision

Evidence metadata shall be a primary retrieval input.

Metadata Dimensions

- Classification
- Domain
- Source
- Trust level
- Governance status
- Retention status

Rationale

Supports efficient and explainable retrieval.

Consequences

Positive

- Improved retrieval accuracy

Negative

- Metadata quality dependency

11. Architectural Decision ADR-0047

Decision Title

Governance-Aware Retrieval Filtering

Status

Approved

Decision

Retrieval results shall be filtered according to governance policies.

Filtering Examples

- Restricted evidence exclusion
- Retention restrictions
- Classification restrictions
- Jurisdiction restrictions

Rationale

Prevents unauthorized evidence usage.

Consequences

Positive

- Strong governance enforcement

Negative

- Additional filtering complexity

12. Architectural Decision ADR-0048

Decision Title

Retrieval Result Traceability

Status

Approved

Decision

Every retrieval operation shall produce traceable retrieval records.

Required Elements

- Request identifier
- Opportunity identifier
- Retrieval criteria
- Retrieved evidence identifiers
- Timestamp
- Requestor identity

Rationale

Supports explainability and audits.

Consequences

Positive

- Complete retrieval visibility

Negative

- Additional storage requirements

13. Architectural Decision ADR-0049

Decision Title

Replay-Compatible Retrieval Reconstruction

Status

Approved

Decision

Retrieval operations shall be reproducible during replay execution.

Required Preservation

- Retrieval criteria
- Retrieval strategy
- Governance rules
- Authority context
- Evidence versions

Rationale

Supports historical reconstruction.

Consequences

Positive

- Accurate replay capability

Negative

- Additional retention obligations

14. Architectural Decision ADR-0050

Decision Title

Retrieval Result Immutability

Status

Approved

Decision

Historical retrieval results shall be preserved as immutable audit records.

Rationale

Protects audit integrity and investigation capabilities.

Consequences

Positive

- Strong audit support

Negative

- Increased retention volume

15. Architectural Decision ADR-0051

Decision Title

Retrieval Explainability Framework

Status

Approved

Decision

The Retrieval Module shall provide explainable retrieval outcomes.

Explainability Elements

- Retrieval strategy used
- Matching criteria
- Filtering rationale
- Governance constraints applied

Rationale

Supports platform transparency objectives.

Consequences

Positive

- Increased trust
- Better governance visibility

Negative

- Additional processing requirements

16. Architectural Decision ADR-0052

Decision Title

Scalable Retrieval Processing

Status

Approved

Decision

Retrieval architecture shall support horizontal growth of retrieval workloads.

Scalability Targets

- Growing evidence volumes
- Increasing opportunities
- Increasing concurrent requests
- Future enterprise expansion

Rationale

Supports long-term platform growth.

Consequences

Positive

- Enterprise scalability

Negative

- Infrastructure planning requirements

17. Architectural Decision ADR-0053

Decision Title

Retrieval Observability Requirements

Status

Approved

Decision

Retrieval operations shall generate operational telemetry.

Required Observability Elements

- Latency
- Throughput
- Success rate
- Error rate
- Resource utilization

Rationale

Supports operational reliability.

Consequences

Positive

- Improved monitoring

Negative

- Additional telemetry storage

18. Architectural Constraints

The Retrieval Module shall comply with:

RET-CON-001

Preserve evidence traceability.

RET-CON-002

Enforce authority validation.

RET-CON-003

Support governance filtering.

RET-CON-004

Maintain retrieval explainability.

RET-CON-005

Support replay reconstruction.

RET-CON-006

Maintain module independence.

RET-CON-007

Preserve audit visibility.

RET-CON-008

Support enterprise scalability.

19. Metrics

The following metrics shall be monitored:

MET-RET-001

Retrieval request volume.

MET-RET-002

Retrieval latency.

MET-RET-003

Retrieval success rate.

MET-RET-004

Evidence retrieval coverage.

MET-RET-005

Governance filter application rate.

MET-RET-006

Retrieval traceability completeness.

MET-RET-007

Replay reconstruction accuracy.

MET-RET-008

Retrieval explainability coverage.

20. Audit Requirements

Audits shall verify:

- Retrieval traceability
- Governance filter enforcement
- Authority validation compliance
- Retrieval criteria preservation
- Retrieval record integrity
- Explainability completeness
- Replay readiness
- Observability coverage

Audit evidence shall be retained according to enterprise retention policies.

21. Compliance Requirements

The Retrieval Module architecture shall support:

- SOC 2 access governance requirements
- ISO 27001 access control obligations
- ISO 27001 information retrieval controls
- Enterprise audit requirements
- Customer security assessments
- Regulatory review activities where applicable
- Internal governance standards

Compliance mappings shall be maintained through RTM, SEC, GOV, AUD, and COMP artifacts.

22. Section Summary

This section defines the approved architecture decisions governing the Retrieval Module, including retrieval strategy abstraction, authority-validated retrieval, governance-aware filtering, metadata-driven retrieval, retrieval traceability, explainability requirements, replay support, scalability considerations, observability controls, audit obligations, and compliance requirements. These decisions establish Retrieval as the governed evidence access layer of TODP and ensure that evidence selection remains transparent, traceable, auditable, explainable, and reproducible throughout the decision lifecycle.

ADR V1 – Architecture Decision Record Repository

Section 8: Ranking Module Architecture Decisions

Document ID: ADR-V1-S08

Document Title: Ranking Module Architecture Decisions

Version: 1.0
Status: Approved
Owner: Enterprise Architecture Function
Review Authority: Architecture Review Board (ARB)
Approval Authority: Product Governance Council
Classification: Internal Confidential

1. Purpose

The purpose of this section is to formally document the architectural decisions governing the Ranking Module within the Transparent Opportunity Decision Platform (TODP).

The Ranking Module is responsible for evaluating retrieved evidence, applying ranking methodologies, calculating opportunity relevance and priority scores, generating ranking outputs, and producing explainable ranking results that support downstream recommendation generation.

The Ranking Module transforms retrieved evidence into ordered, scored, and explainable evaluation outcomes while maintaining transparency, governance compliance, auditability, and replayability.

2. Objectives

OBJ-RNK-001

Provide objective and explainable ranking outcomes.

OBJ-RNK-002

Support evidence-based prioritization.

OBJ-RNK-003

Maintain ranking transparency.

OBJ-RNK-004

Enable governance oversight of ranking methodologies.

OBJ-RNK-005

Support scalable ranking operations.

OBJ-RNK-006

Preserve ranking traceability.

OBJ-RNK-007

Enable ranking replay reconstruction.

OBJ-RNK-008

Support future ranking methodology evolution.

3. Scope

The Ranking Module architecture governs:

- Ranking input processing
- Ranking model execution
- Score generation
- Opportunity prioritization
- Ranking explanations
- Ranking traceability
- Ranking governance controls
- Ranking version management
- Ranking replay support

The module operates within the TODP workflow as follows:

Opportunity

→ Authority

→ Evidence

→ Retrieval

→ Ranking

→ Recommendation

→ Governance

→ Audit

→ Replay

4. Architectural Decision ADR-0054

Decision Title

Ranking as Independent Domain Module

Status

Approved

Decision

The Ranking Module shall operate as an independent architectural domain with dedicated ownership, governance controls, version management, and lifecycle oversight.

Rationale

Ranking methodologies evolve independently from retrieval, recommendation, and governance functions.

Consequences

Positive

- Independent evolution
- Reduced coupling
- Clear ownership

Negative

- Additional interface management

5. Architectural Decision ADR-0055

Decision Title

Evidence-Driven Ranking Architecture

Status

Approved

Decision

Ranking decisions shall be based exclusively on evidence supplied through approved Retrieval processes.

Rationale

Maintains transparency and prevents unsupported ranking outcomes.

Consequences

Positive

- Improved trust
- Better explainability

Negative

- Dependency on retrieval quality

6. Architectural Decision ADR-0056

Decision Title

Opportunity-Centric Ranking Context

Status

Approved

Decision

All ranking evaluations shall execute within the context of a specific Opportunity.

Required Inputs

- Opportunity identifier
- Retrieved evidence set
- Ranking configuration
- Governance constraints

Rationale

Ensures contextual consistency.

Consequences

Positive

- Improved relevance

Negative

- Additional contextual dependencies

7. Architectural Decision ADR-0057

Decision Title

Ranking Methodology Abstraction Layer

Status

Approved

Decision

Ranking logic shall be abstracted from ranking methodology implementations.

Supported Categories

- Rule-based ranking
- Weighted ranking
- Statistical ranking
- Hybrid ranking
- Future approved methodologies

Rationale

Supports future innovation without redesigning platform consumers.

Consequences

Positive

- Extensibility
- Reduced architectural rigidity

Negative

- Additional abstraction complexity

8. Architectural Decision ADR-0058

Decision Title

Explainable Ranking Requirement

Status

Approved

Decision

All ranking outcomes shall provide explainable reasoning.

Required Explanation Components

- Ranking inputs
- Evidence references
- Scoring factors
- Weighting contributions
- Ranking rationale

Rationale

Supports transparency and governance objectives.

Consequences

Positive

- Increased stakeholder trust

Negative

- Additional processing overhead

9. Architectural Decision ADR-0059

Decision Title

Versioned Ranking Models

Status

Approved

Decision

Ranking methodologies shall be version controlled.

Version Requirements

- Unique version identifiers
- Approval history
- Effective dates
- Change history
- Governance approvals

Rationale

Supports auditability and replay reconstruction.

Consequences

Positive

- Controlled evolution

Negative

- Version management effort

10. Architectural Decision ADR-0060

Decision Title

Governance-Controlled Ranking Configuration

Status

Approved

Decision

Ranking configuration changes shall be governed through approved change controls.

Governed Elements

- Scoring rules
- Weighting factors
- Thresholds
- Ranking criteria
- Prioritization logic

Rationale

Prevents uncontrolled ranking behavior changes.

Consequences

Positive

- Strong governance oversight

Negative

- Increased administrative controls

11. Architectural Decision ADR-0061

Decision Title

Ranking Input Traceability

Status

Approved

Decision

All ranking inputs shall maintain traceable references to source evidence.

Required Traceability Elements

- Evidence identifiers
- Retrieval identifiers
- Opportunity identifiers
- Authority context

Rationale

Supports audit investigations and explainability.

Consequences

Positive

- Complete lineage visibility

Negative

- Additional metadata requirements

12. Architectural Decision ADR-0062

Decision Title

Ranking Output Preservation

Status

Approved

Decision

Ranking outputs shall be preserved as immutable evaluation records.

Preserved Elements

- Score outputs
- Ranking order
- Explanations
- Configuration versions
- Input references

Rationale

Supports audit and replay requirements.

Consequences

Positive

- Strong historical visibility

Negative

- Increased storage utilization

13. Architectural Decision ADR-0063

Decision Title

Replay-Compatible Ranking Execution

Status

Approved

Decision

Ranking operations shall be reproducible during replay activities.

Required Preservation

- Input evidence versions
- Ranking configuration versions
- Governance constraints
- Execution context

Rationale

Supports reconstruction of historical outcomes.

Consequences

Positive

- Replay accuracy

Negative

- Long-term retention requirements

14. Architectural Decision ADR-0064

Decision Title

Ranking Governance Validation

Status

Approved

Decision

Ranking outputs shall be evaluated against governance policies before downstream recommendation processing.

Validation Examples

- Policy thresholds
- Regulatory restrictions
- Governance exclusions
- Risk controls

Rationale

Ensures ranking compliance.

Consequences

Positive

- Improved governance integrity

Negative

- Additional validation operations

15. Architectural Decision ADR-0065

Decision Title

Ranking Quality Monitoring

Status

Approved

Decision

The Ranking Module shall continuously monitor ranking quality indicators.

Quality Indicators

- Ranking consistency
- Ranking stability
- Score distribution
- Configuration integrity
- Governance compliance

Rationale

Supports operational quality management.

Consequences

Positive

- Improved reliability

Negative

- Additional monitoring overhead

16. Architectural Decision ADR-0066

Decision Title

Scalable Ranking Processing

Status

Approved

Decision

Ranking architecture shall support increasing evaluation volumes without architectural redesign.

Scalability Targets

- Growing evidence volumes
- Growing opportunity volumes
- Growing concurrent evaluations
- Future enterprise expansion

Rationale

Supports long-term growth.

Consequences

Positive

- Enterprise scalability

Negative

- Infrastructure planning complexity

17. Architectural Decision ADR-0067

Decision Title

Ranking Observability Framework

Status

Approved

Decision

The Ranking Module shall generate operational telemetry supporting monitoring and diagnostics.

Required Telemetry

- Evaluation volume
- Latency
- Error rates
- Configuration usage
- Resource utilization

Rationale

Supports operational reliability.

Consequences

Positive

- Improved visibility

Negative

- Additional telemetry management

18. Architectural Constraints

The Ranking Module shall comply with:

RNK-CON-001

Maintain ranking explainability.

RNK-CON-002

Preserve evidence traceability.

RNK-CON-003

Support governance validation.

RNK-CON-004

Maintain configuration versioning.

RNK-CON-005

Support replay reconstruction.

RNK-CON-006

Preserve audit visibility.

RNK-CON-007

Maintain module independence.

RNK-CON-008

Support enterprise scalability.

19. Metrics

The following metrics shall be monitored:

MET-RNK-001

Ranking execution volume.

MET-RNK-002

Ranking latency.

MET-RNK-003

Ranking success rate.

MET-RNK-004

Ranking explanation coverage.

MET-RNK-005

Governance validation pass rate.

MET-RNK-006

Ranking traceability completeness.

MET-RNK-007

Replay ranking accuracy.

MET-RNK-008

Ranking configuration compliance rate.

20. Audit Requirements

Audits shall verify:

- Ranking methodology approvals
- Ranking version management
- Input traceability completeness
- Output preservation integrity

- Governance validation compliance
- Explainability coverage
- Replay readiness
- Operational monitoring coverage

Audit evidence shall be retained according to enterprise retention policies.

21. Compliance Requirements

The Ranking Module architecture shall support:

- SOC 2 governance controls
- ISO 27001 change management controls
- ISO 27001 information integrity controls
- Enterprise audit requirements
- Customer due diligence reviews
- Regulatory review obligations where applicable
- Internal governance standards

Compliance mappings shall be maintained through RTM, SEC, GOV, AUD, and COMP artifacts.

22. Section Summary

This section defines the approved architecture decisions governing the Ranking Module, including evidence-driven ranking, methodology abstraction, explainability requirements, version-controlled ranking models, governance validation, input traceability, output preservation, replay support, observability controls, audit obligations, and compliance requirements. These decisions establish Ranking as the transparent evaluation layer of TODP and ensure that prioritization outcomes remain explainable, traceable, governed, auditable, and reproducible throughout the decision lifecycle.

ADR V1 – Architecture Decision Record Repository

Section 9: Recommendation Module Architecture Decisions

Document ID: ADR-V1-S09

Document Title: Recommendation Module Architecture Decisions

Version: 1.0

Status: Approved

Owner: Enterprise Architecture Function

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to formally document the architectural decisions governing the Recommendation Module within the Transparent Opportunity Decision Platform (TODP).

The Recommendation Module is responsible for transforming ranked opportunity evaluations into explainable recommendations that support human decision-makers. The module generates recommendation outcomes, recommendation rationales, recommendation confidence indicators, and recommendation traceability artifacts while preserving transparency, governance, auditability, and replayability.

The Recommendation Module is a decision-support capability and shall not function as an autonomous decision-making authority.

2. Objectives

OBJ-REC-001

Generate transparent recommendations.

OBJ-REC-002

Maintain explainable recommendation outcomes.

OBJ-REC-003

Support evidence-based decision support.

OBJ-REC-004

Preserve human oversight.

OBJ-REC-005

Enable governance review of recommendations.

OBJ-REC-006

Support audit and compliance requirements.

OBJ-REC-007

Enable replay reconstruction.

OBJ-REC-008

Support future recommendation methodology evolution.

3. Scope

The Recommendation Module architecture governs:

- Recommendation generation
- Recommendation scoring
- Recommendation rationale generation

- Recommendation confidence assessment
- Recommendation traceability
- Recommendation governance validation
- Recommendation version management
- Recommendation replay support

The module operates within the TODP workflow as follows:

Opportunity

- Authority
- Evidence
- Retrieval
- Ranking
- Recommendation
- Governance
- Audit
- Replay

4. Architectural Decision ADR-0068

Decision Title

Recommendation as Independent Domain Module

Status

Approved

Decision

The Recommendation Module shall operate as an independent architectural domain with dedicated ownership, governance controls, lifecycle management, and audit requirements.

Rationale

Recommendation logic evolves independently from ranking methodologies and governance workflows.

Consequences

Positive

- Independent evolution
- Reduced coupling
- Clear accountability

Negative

- Additional integration management

5. Architectural Decision ADR-0069

Decision Title

Decision Support Architecture

Status

Approved

Decision

The Recommendation Module shall function exclusively as a decision-support capability and shall not perform autonomous final decision-making.

Rationale

TODP is designed to support transparent decision-making rather than replace accountable human authority.

Consequences

Positive

- Strong governance alignment
- Reduced regulatory risk
- Improved accountability

Negative

- Human review dependency

6. Architectural Decision ADR-0070

Decision Title

Ranking-Driven Recommendation Inputs

Status

Approved

Decision

Recommendation generation shall be based on approved ranking outputs and supporting evidence references.

Required Inputs

- Opportunity identifier
- Ranking results
- Evidence references
- Governance constraints

Rationale

Maintains consistency between evaluation and recommendation activities.

Consequences

Positive

- Strong traceability
- Better explainability

Negative

- Dependency on ranking quality

7. Architectural Decision ADR-0071

Decision Title

Recommendation Methodology Abstraction Layer

Status

Approved

Decision

Recommendation generation logic shall be abstracted from recommendation methodologies.

Supported Methodology Categories

- Rule-based recommendations
- Weighted recommendations
- Policy-driven recommendations
- Hybrid recommendation strategies
- Future approved methodologies

Rationale

Supports long-term adaptability.

Consequences

Positive

- Extensibility
- Future readiness

Negative

- Additional abstraction complexity

8. Architectural Decision ADR-0072

Decision Title

Explainable Recommendation Requirement

Status

Approved

Decision

All recommendations shall include explainable rationale information.

Required Components

- Recommendation outcome
- Supporting evidence
- Ranking factors
- Decision rationale
- Governance considerations

Rationale

Supports transparency and stakeholder trust.

Consequences

Positive

- Increased confidence
- Improved review quality

Negative

- Additional processing requirements

9. Architectural Decision ADR-0073

Decision Title

Recommendation Confidence Framework

Status

Approved

Decision

Recommendations shall include confidence indicators derived from available evidence and ranking outputs.

Confidence Factors

- Evidence completeness
- Evidence quality
- Ranking consistency
- Governance compliance
- Data freshness

Rationale

Provides context for human reviewers.

Consequences

Positive

- Better decision support

Negative

- Additional calculation complexity

10. Architectural Decision ADR-0074

Decision Title

Version-Controlled Recommendation Models

Status

Approved

Decision

Recommendation methodologies shall be maintained under version control.

Version Requirements

- Version identifier
- Approval history
- Change history

- Effective date
- Governance approval records

Rationale

Supports auditability and replay.

Consequences

Positive

- Controlled evolution

Negative

- Version management overhead

11. Architectural Decision ADR-0075

Decision Title

Recommendation Output Preservation

Status

Approved

Decision

Recommendation outputs shall be preserved as immutable historical records.

Preserved Elements

- Recommendation outcome
- Recommendation rationale
- Confidence indicators
- Input references
- Methodology version

Rationale

Supports audits and historical investigations.

Consequences

Positive

- Strong historical integrity

Negative

- Additional storage requirements

12. Architectural Decision ADR-0076

Decision Title

Human Review Enforcement

Status

Approved

Decision

Recommendation outputs shall be subject to human review before governance approval and final action.

Review Requirements

- Reviewer identification
- Review outcome
- Review timestamp
- Review rationale

Rationale

Preserves human accountability.

Consequences

Positive

- Reduced automation risk
- Improved governance

Negative

- Additional review effort

13. Architectural Decision ADR-0077

Decision Title

Governance-Aware Recommendation Validation

Status

Approved

Decision

Recommendations shall be validated against governance policies before progressing to Governance Review.

Validation Categories

- Policy compliance
- Regulatory constraints
- Risk thresholds
- Governance restrictions

Rationale

Ensures recommendations remain within approved governance boundaries.

Consequences

Positive

- Strong governance integrity

Negative

- Additional validation complexity

14. Architectural Decision ADR-0078

Decision Title

Recommendation Traceability Framework

Status

Approved

Decision

Every recommendation shall maintain complete traceability to upstream inputs.

Traceability Elements

- Opportunity identifier
- Authority context
- Evidence references
- Retrieval records
- Ranking outputs
- Governance validations

Rationale

Supports transparency and auditability.

Consequences

Positive

- Complete lineage visibility

Negative

- Increased metadata management

15. Architectural Decision ADR-0079

Decision Title

Replay-Compatible Recommendation Reconstruction

Status

Approved

Decision

Recommendation outcomes shall be reproducible during replay activities.

Required Preservation

- Recommendation version
- Ranking version
- Evidence versions
- Governance rules
- Execution context

Rationale

Supports accurate historical reconstruction.

Consequences

Positive

- Replay integrity

Negative

- Long-term retention requirements

16. Architectural Decision ADR-0080

Decision Title

Recommendation Observability Framework

Status

Approved

Decision

Recommendation processing shall generate operational telemetry.

Required Telemetry

- Recommendation volume
- Processing latency
- Error rates
- Methodology usage
- Confidence distribution

Rationale

Supports operational reliability.

Consequences

Positive

- Improved monitoring

Negative

- Additional telemetry storage

17. Architectural Decision ADR-0081

Decision Title

Scalable Recommendation Processing

Status

Approved

Decision

Recommendation architecture shall support increasing evaluation volumes without requiring fundamental redesign.

Scalability Targets

- Growing opportunity volume
- Growing evidence volume
- Growing user activity
- Enterprise-scale adoption

Rationale

Supports long-term growth objectives.

Consequences

Positive

- Enterprise scalability

Negative

- Infrastructure planning complexity

18. Architectural Constraints

The Recommendation Module shall comply with:

REC-CON-001

Maintain recommendation explainability.

REC-CON-002

Preserve human oversight.

REC-CON-003

Support governance validation.

REC-CON-004

Maintain recommendation traceability.

REC-CON-005

Support replay reconstruction.

REC-CON-006

Preserve audit visibility.

REC-CON-007

Maintain module independence.

REC-CON-008

Support enterprise scalability.

19. Metrics

The following metrics shall be monitored:

MET-REC-001

Recommendation generation volume.

MET-REC-002

Recommendation latency.

MET-REC-003

Recommendation success rate.

MET-REC-004

Recommendation explanation coverage.

MET-REC-005

Human review completion rate.

MET-REC-006

Governance validation pass rate.

MET-REC-007

Recommendation traceability completeness.

MET-REC-008

Replay recommendation accuracy.

20. Audit Requirements

Audits shall verify:

- Recommendation methodology approvals
- Human review enforcement
- Recommendation traceability
- Governance validation compliance
- Recommendation output integrity
- Explainability completeness

- Replay readiness
- Observability coverage

Audit evidence shall be retained according to enterprise retention policies.

21. Compliance Requirements

The Recommendation Module architecture shall support:

- SOC 2 governance controls
- ISO 27001 change management requirements
- ISO 27001 information integrity controls
- Enterprise audit requirements
- Customer security reviews
- Regulatory assessment activities where applicable
- Internal governance standards

Compliance mappings shall be maintained through RTM, SEC, GOV, AUD, and COMP artifacts.

22. Section Summary

This section defines the approved architecture decisions governing the Recommendation Module, including decision-support architecture, recommendation methodology abstraction, explainability requirements, confidence assessment, human review enforcement, governance validation, traceability controls, output preservation, replay support, observability requirements, audit obligations, and compliance requirements. These decisions establish Recommendation as the explainable decision-support layer of TODP and ensure that recommendation outcomes remain transparent, governed, traceable, auditable, human-supervised, and reproducible throughout the decision lifecycle.

ADR V1 – Architecture Decision Record Repository

Section 10: Governance Module Architecture Decisions

Document ID: ADR-V1-S10

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Owner: Enterprise Architecture Function

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to formally document the architectural decisions governing the Governance Module within the Transparent Opportunity Decision Platform (TODP).

The Governance Module is responsible for policy enforcement, decision oversight, exception management, approval workflows, compliance validation, risk control enforcement, and governance accountability across the TODP decision lifecycle.

The Governance Module serves as the control layer ensuring that recommendations, decisions, and actions remain aligned with enterprise policies, regulatory obligations, risk tolerances, and organizational governance requirements.

2. Objectives

OBJ-GOV-001

Provide centralized governance oversight.

OBJ-GOV-002

Enforce enterprise policies consistently.

OBJ-GOV-003

Support accountable decision-making.

OBJ-GOV-004

Enable governance transparency.

OBJ-GOV-005

Maintain compliance alignment.

OBJ-GOV-006

Support audit and investigation activities.

OBJ-GOV-007

Enable governance replay reconstruction.

OBJ-GOV-008

Support scalable governance operations.

3. Scope

The Governance Module architecture governs:

- Policy enforcement
- Governance reviews
- Approval workflows
- Exception management

- Risk validation
- Compliance validation
- Governance decisions
- Governance traceability
- Governance reporting
- Governance replay support

The module operates within the TODP workflow as follows:

Opportunity

→ Authority

→ Evidence

→ Retrieval

→ Ranking

→ Recommendation

→ Governance

→ Audit

→ Replay

4. Architectural Decision ADR-0082

Decision Title

Governance as Independent Domain Module

Status

Approved

Decision

The Governance Module shall operate as an independent architectural domain with dedicated ownership, governance authorities, lifecycle management, and operational controls.

Rationale

Governance responsibilities extend across all platform modules and require independent oversight.

Consequences

Positive

- Clear accountability
- Reduced conflicts of interest
- Independent evolution

Negative

- Additional coordination requirements

5. Architectural Decision ADR-0083

Decision Title

Centralized Governance Enforcement Architecture

Status

Approved

Decision

Policy enforcement shall be centralized within the Governance Module rather than distributed across individual business modules.

Rationale

Ensures consistent policy interpretation and enforcement.

Consequences

Positive

- Uniform governance behavior
- Simplified policy management

Negative

- Centralized dependency

6. Architectural Decision ADR-0084

Decision Title

Governance Review Before Final Decision Execution

Status

Approved

Decision

Recommendations requiring formal approval shall undergo governance review prior to final execution.

Governance Inputs

- Opportunity context
- Authority validation
- Evidence references

- Ranking outputs
- Recommendation outputs

Rationale

Provides independent governance oversight.

Consequences

Positive

- Reduced governance risk
- Improved accountability

Negative

- Additional review activities

7. Architectural Decision ADR-0085

Decision Title

Policy-Driven Governance Model

Status

Approved

Decision

Governance decisions shall be based on formally approved and version-controlled policies.

Policy Categories

- Business policies
- Risk policies
- Security policies
- Compliance policies
- Operational policies
- Governance policies

Rationale

Ensures predictable and repeatable governance behavior.

Consequences

Positive

- Consistency
- Transparency

Negative

- Policy administration effort

8. Architectural Decision ADR-0086

Decision Title

Governance Policy Repository

Status

Approved

Decision

All governance policies shall be maintained within a centralized governance policy repository.

Repository Requirements

- Version control
- Approval history
- Effective dates
- Change tracking
- Audit records

Rationale

Supports policy integrity and governance traceability.

Consequences

Positive

- Centralized management

Negative

- Repository maintenance overhead

9. Architectural Decision ADR-0087

Decision Title

Governance Decision Traceability

Status

Approved

Decision

All governance decisions shall maintain complete traceability.

Required Traceability Elements

- Opportunity identifier
- Policy references
- Governance reviewer
- Review outcome
- Decision rationale
- Timestamp

Rationale

Supports accountability and auditability.

Consequences

Positive

- Full governance visibility

Negative

- Additional metadata requirements

10. Architectural Decision ADR-0088

Decision Title

Exception Management Framework

Status

Approved

Decision

The Governance Module shall provide controlled exception management capabilities.

Exception Requirements

- Documented justification
- Approval workflow
- Risk assessment

- Expiration criteria
- Audit tracking

Rationale

Supports controlled operational flexibility.

Consequences

Positive

- Managed policy exceptions

Negative

- Additional governance administration

11. Architectural Decision ADR-0089

Decision Title

Risk-Based Governance Validation

Status

Approved

Decision

Governance reviews shall evaluate risk implications before approval decisions.

Risk Categories

- Operational risk
- Security risk
- Compliance risk
- Business risk
- Reputational risk

Rationale

Supports informed governance decisions.

Consequences

Positive

- Improved risk management

Negative

- Additional evaluation effort

12. Architectural Decision ADR-0090

Decision Title

Separation of Governance and Recommendation Functions

Status

Approved

Decision

Governance reviewers shall remain independent from recommendation generation activities.

Rationale

Reduces bias and conflict-of-interest risk.

Consequences

Positive

- Strong governance integrity

Negative

- Additional role management

13. Architectural Decision ADR-0091

Decision Title

Version-Controlled Governance Rules

Status

Approved

Decision

Governance rules and policy logic shall be version controlled.

Version Requirements

- Unique identifiers
- Effective dates
- Approval history
- Revision history

Rationale

Supports auditability and replay reconstruction.

Consequences

Positive

- Controlled governance evolution

Negative

- Version management effort

14. Architectural Decision ADR-0092

Decision Title

Governance Outcome Preservation

Status

Approved

Decision

Governance outcomes shall be preserved as immutable records.

Preserved Elements

- Review results
- Approval decisions
- Policy references
- Reviewer identity
- Supporting rationale

Rationale

Supports audits and investigations.

Consequences

Positive

- Strong historical integrity

Negative

- Storage overhead

15. Architectural Decision ADR-0093

Decision Title

Replay-Compatible Governance Reconstruction

Status

Approved

Decision

Governance decisions shall be reproducible during replay operations.

Required Preservation

- Policy versions
- Governance rules
- Review context
- Risk assessments
- Governance outcomes

Rationale

Supports accurate historical reconstruction.

Consequences

Positive

- Replay accuracy

Negative

- Long-term retention obligations

16. Architectural Decision ADR-0094

Decision Title

Governance Reporting Framework

Status

Approved

Decision

The Governance Module shall generate governance reporting artifacts.

Reporting Categories

- Approval statistics
- Exception reports
- Policy compliance reports
- Governance activity reports
- Risk reports

Rationale

Supports oversight and accountability.

Consequences

Positive

- Improved governance visibility

Negative

- Reporting maintenance effort

17. Architectural Decision ADR-0095

Decision Title

Governance Observability Framework

Status

Approved

Decision

Governance operations shall generate operational telemetry.

Required Telemetry

- Review volume
- Approval volume
- Exception volume
- Policy usage
- Processing latency

Rationale

Supports operational management.

Consequences

Positive

- Better monitoring

Negative

- Additional telemetry requirements

18. Architectural Decision ADR-0096

Decision Title

Scalable Governance Processing

Status

Approved

Decision

Governance architecture shall support increasing review volumes and policy complexity without fundamental redesign.

Scalability Targets

- Growing opportunities
- Growing recommendations
- Growing governance reviews
- Enterprise-scale adoption

Rationale

Supports future growth.

Consequences

Positive

- Enterprise scalability

Negative

- Capacity planning requirements

19. Architectural Constraints

The Governance Module shall comply with:

GOV-CON-001

Maintain policy-driven enforcement.

GOV-CON-002

Preserve reviewer independence.

GOV-CON-003

Support exception management.

GOV-CON-004

Maintain governance traceability.

GOV-CON-005

Support replay reconstruction.

GOV-CON-006

Preserve audit visibility.

GOV-CON-007

Maintain version-controlled policies.

GOV-CON-008

Support enterprise scalability.

20. Metrics

The following metrics shall be monitored:

MET-GOV-001

Governance review volume.

MET-GOV-002

Approval rate.

MET-GOV-003

Exception rate.

MET-GOV-004

Policy compliance rate.

MET-GOV-005

Governance processing latency.

MET-GOV-006

Governance traceability completeness.

MET-GOV-007

Replay governance accuracy.

MET-GOV-008

Governance policy utilization rate.

21. Audit Requirements

Audits shall verify:

- Policy enforcement consistency
- Governance independence
- Exception management compliance
- Governance traceability
- Policy version management
- Outcome preservation integrity
- Replay readiness
- Reporting completeness

Audit evidence shall be retained according to enterprise retention policies.

22. Compliance Requirements

The Governance Module architecture shall support:

- SOC 2 governance controls
- ISO 27001 governance requirements
- ISO 27001 risk management controls
- Enterprise audit requirements
- Customer due diligence reviews
- Regulatory oversight obligations where applicable
- Internal governance standards

Compliance mappings shall be maintained through RTM, SEC, GOV, AUD, COMP, and REL artifacts.

23. Section Summary

This section defines the approved architecture decisions governing the Governance Module, including centralized policy enforcement, governance review workflows, exception

management, risk-based validation, governance traceability, policy version control, outcome preservation, reporting capabilities, replay support, observability requirements, audit obligations, and compliance requirements. These decisions establish Governance as the independent control layer of TODP and ensure that decision outcomes remain policy-compliant, accountable, traceable, auditable, explainable, and reproducible throughout the platform lifecycle.

ADR V1 – Architecture Decision Record Repository

Section 11: Audit Module Architecture Decisions

Document ID: ADR-V1-S11

Document Title: Audit Module Architecture Decisions

Version: 1.0

Status: Approved

Owner: Enterprise Architecture Function

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to formally document the architectural decisions governing the Audit Module within the Transparent Opportunity Decision Platform (TODP).

The Audit Module is responsible for capturing, preserving, correlating, validating, and presenting audit evidence generated across the TODP platform. The module serves as the authoritative audit capability supporting investigations, compliance reviews, governance oversight, forensic analysis, operational reviews, customer audits, and regulatory assessments.

The Audit Module ensures that all platform activities remain observable, traceable, reviewable, and historically reconstructable.

2. Objectives

OBJ-AUD-001

Provide centralized audit evidence management.

OBJ-AUD-002

Maintain immutable audit records.

OBJ-AUD-003

Support enterprise audit requirements.

OBJ-AUD-004

Enable investigation and forensic analysis.

OBJ-AUD-005

Support compliance validation.

OBJ-AUD-006

Preserve end-to-end traceability.

OBJ-AUD-007

Enable replay reconstruction.

OBJ-AUD-008

Support long-term audit retention.

3. Scope

The Audit Module architecture governs:

- Audit event collection
- Audit evidence preservation
- Audit record management
- Audit correlation
- Audit reporting
- Audit search and retrieval
- Investigation support
- Compliance evidence management
- Replay support

The module operates across all stages of the TODP workflow:

Opportunity

→ Authority

→ Evidence

→ Retrieval

→ Ranking

→ Recommendation

→ Governance

→ Audit

→ Replay

4. Architectural Decision ADR-0097

Decision Title

Audit as Independent Domain Module

Status

Approved

Decision

The Audit Module shall operate as an independent architectural domain with dedicated ownership, governance controls, retention requirements, and operational oversight.

Rationale

Audit responsibilities extend across all platform components and require independent control.

Consequences

Positive

- Independent oversight
- Reduced conflict of interest
- Improved audit integrity

Negative

- Additional integration requirements

5. Architectural Decision ADR-0098

Decision Title

Centralized Audit Collection Architecture

Status

Approved

Decision

Audit events shall be collected through a centralized audit collection capability.

Sources

- Opportunity Module
- Authority Module
- Evidence Module
- Retrieval Module
- Ranking Module
- Recommendation Module
- Governance Module
- Replay Module
- Platform Services

Rationale

Ensures consistent audit coverage.

Consequences

Positive

- Uniform audit standards
- Simplified investigations

Negative

- Centralized dependency

6. Architectural Decision ADR-0099

Decision Title

Immutable Audit Record Preservation

Status

Approved

Decision

Audit records shall be preserved as immutable historical artifacts.

Requirements

- No destructive modification
- Historical integrity preservation
- Tamper-evident controls
- Retention enforcement

Rationale

Supports audit trustworthiness.

Consequences

Positive

- Strong evidentiary value

Negative

- Additional storage requirements

7. Architectural Decision ADR-0100

Decision Title

Comprehensive Audit Event Coverage

Status

Approved

Decision

All material platform activities shall generate audit records.

Minimum Audit Categories

- Authentication events
- Authorization events
- Opportunity events
- Evidence events
- Retrieval events
- Ranking events
- Recommendation events
- Governance events
- Replay events
- Administrative events

Rationale

Supports complete platform visibility.

Consequences

Positive

- Broad audit coverage

Negative

- Increased audit volume

8. Architectural Decision ADR-0101

Decision Title

Audit Correlation Framework

Status

Approved

Decision

Audit records shall support end-to-end correlation across the TODP workflow.

Correlation Attributes

- Opportunity ID
- Authority ID
- Evidence ID
- Retrieval ID
- Ranking ID
- Recommendation ID
- Governance ID
- Replay ID

Rationale

Supports investigations and traceability.

Consequences

Positive

- End-to-end visibility

Negative

- Metadata management effort

9. Architectural Decision ADR-0102

Decision Title

Audit Evidence Repository

Status

Approved

Decision

Audit records shall be stored within a dedicated audit evidence repository.

Repository Requirements

- Controlled access
- Retention enforcement
- Searchability
- Integrity protection
- Scalability

Rationale

Supports centralized audit management.

Consequences

Positive

- Simplified governance

Negative

- Repository administration effort

10. Architectural Decision ADR-0103

Decision Title

Audit Search and Investigation Support

Status

Approved

Decision

The Audit Module shall provide capabilities supporting investigations and audit reviews.

Capabilities

- Search
- Filtering
- Correlation
- Timeline reconstruction
- Evidence export

Rationale

Supports operational and compliance investigations.

Consequences

Positive

- Faster investigations

Negative

- Additional operational complexity

11. Architectural Decision ADR-0104

Decision Title

Audit Access Governance

Status

Approved

Decision

Audit information access shall be governed through Authority Module controls and governance policies.

Access Categories

- Auditor access
- Compliance access
- Governance access
- Administrative access

Rationale

Protects sensitive audit information.

Consequences

Positive

- Improved security

Negative

- Additional access management

12. Architectural Decision ADR-0105

Decision Title

Audit Retention Governance

Status

Approved

Decision

Audit records shall follow formally approved retention policies.

Retention Requirements

- Retention classification
- Legal hold support
- Archive support
- Controlled disposition

Rationale

Supports regulatory and enterprise obligations.

Consequences

Positive

- Compliance readiness

Negative

- Retention management overhead

13. Architectural Decision ADR-0106

Decision Title

Replay-Compatible Audit Preservation

Status

Approved

Decision

Audit records shall support replay reconstruction.

Required Preservation

- Historical events
- Historical states
- Governance outcomes
- Evidence references
- Version references

Rationale

Supports accurate replay execution.

Consequences

Positive

- Strong replay support

Negative

- Long-term storage obligations

14. Architectural Decision ADR-0107

Decision Title

Audit Integrity Verification Framework

Status

Approved

Decision

The Audit Module shall implement integrity validation controls.

Validation Categories

- Record integrity
- Chain integrity
- Timestamp integrity
- Retention integrity

Rationale

Protects audit trustworthiness.

Consequences

Positive

- Improved evidentiary confidence

Negative

- Additional validation processing

15. Architectural Decision ADR-0108

Decision Title

Compliance Evidence Management

Status

Approved

Decision

The Audit Module shall support compliance evidence collection and reporting.

Supported Areas

- Security compliance
- Governance compliance
- Operational compliance
- Regulatory compliance

Rationale

Simplifies compliance assessments.

Consequences

Positive

- Better compliance readiness

Negative

- Additional reporting obligations

16. Architectural Decision ADR-0109

Decision Title

Audit Reporting Framework

Status

Approved

Decision

The Audit Module shall generate audit and compliance reports.

Report Categories

- Activity reports
- Compliance reports
- Investigation reports
- Governance reports
- Risk reports

Rationale

Supports oversight and accountability.

Consequences

Positive

- Improved visibility

Negative

- Reporting administration effort

17. Architectural Decision ADR-0110

Decision Title

Audit Observability Framework

Status

Approved

Decision

The Audit Module shall generate telemetry supporting operational monitoring.

Telemetry Categories

- Audit ingestion volume
- Search volume
- Processing latency
- Integrity validation outcomes
- Storage utilization

Rationale

Supports operational management.

Consequences

Positive

- Improved reliability

Negative

- Additional telemetry storage

18. Architectural Decision ADR-0111

Decision Title

Scalable Audit Architecture

Status

Approved

Decision

Audit architecture shall support enterprise-scale growth without fundamental redesign.

Scalability Targets

- Growing event volumes
- Growing audit records
- Growing compliance demands
- Growing investigation workloads

Rationale

Supports long-term enterprise adoption.

Consequences

Positive

- Enterprise scalability

Negative

- Infrastructure planning complexity

19. Architectural Constraints

The Audit Module shall comply with:

AUD-CON-001

Maintain immutable audit records.

AUD-CON-002

Preserve end-to-end traceability.

AUD-CON-003

Support audit correlation.

AUD-CON-004

Enforce retention policies.

AUD-CON-005

Support replay reconstruction.

AUD-CON-006

Protect audit integrity.

AUD-CON-007

Maintain governance-controlled access.

AUD-CON-008

Support enterprise scalability.

20. Metrics

The following metrics shall be monitored:

MET-AUD-001

Audit event volume.

MET-AUD-002

Audit ingestion success rate.

MET-AUD-003

Audit search latency.

MET-AUD-004

Audit integrity validation success rate.

MET-AUD-005

Retention compliance rate.

MET-AUD-006

Audit traceability completeness.

MET-AUD-007

Replay audit reconstruction accuracy.

MET-AUD-008

Compliance evidence coverage.

21. Audit Requirements

Audits of the Audit Module shall verify:

- Audit completeness
- Audit integrity
- Retention compliance
- Access governance compliance
- Correlation capability
- Investigation readiness
- Replay support readiness
- Compliance evidence availability

Audit findings shall be documented and tracked through formal remediation processes.

22. Compliance Requirements

The Audit Module architecture shall support:

- SOC 2 audit logging controls
- SOC 2 monitoring requirements
- ISO 27001 logging controls
- ISO 27001 monitoring controls
- ISO 27001 audit evidence requirements
- Enterprise audit programs
- Customer security assessments
- Regulatory review activities where applicable

Compliance mappings shall be maintained through RTM, SEC, GOV, AUD, COMP, and REL artifacts.

23. Section Summary

This section defines the approved architecture decisions governing the Audit Module, including centralized audit collection, immutable audit preservation, comprehensive audit event coverage, audit correlation, audit evidence management, investigation support, retention governance, integrity validation, compliance reporting, replay support, observability requirements, audit obligations, and compliance requirements. These decisions establish Audit as the authoritative oversight and evidentiary layer of TODP and ensure that all platform activities remain traceable, reviewable, auditable, compliant, and reproducible throughout the platform lifecycle.

ADR V1 – Architecture Decision Record Repository

Section 12: Replay Module Architecture Decisions

Document ID: ADR-V1-S12

Document Title: Replay Module Architecture Decisions

Version: 1.0

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Owner: Enterprise Architecture Function

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to formally document the architectural decisions governing the Replay Module within the Transparent Opportunity Decision Platform (TODP).

The Replay Module is responsible for reconstructing historical decision flows, reproducing prior evaluation outcomes, validating governance actions, supporting audit investigations, enabling forensic analysis, and demonstrating end-to-end decision transparency.

Replay is a foundational TODP capability that differentiates the platform from traditional decision-support systems by allowing stakeholders to reconstruct how a decision was reached using historical evidence, rules, policies, authority states, ranking methodologies, recommendation logic, and governance outcomes.

2. Objectives

OBJ-RPL-001

Enable reconstruction of historical decisions.

OBJ-RPL-002

Support audit investigations.

OBJ-RPL-003

Provide transparent decision replay.

OBJ-RPL-004

Validate governance and compliance outcomes.

OBJ-RPL-005

Support forensic analysis.

OBJ-RPL-006

Preserve end-to-end traceability.

OBJ-RPL-007

Provide reproducible decision outcomes.

OBJ-RPL-008

Support enterprise accountability.

3. Scope

The Replay Module architecture governs:

- Historical decision reconstruction
- Replay execution
- Historical state restoration
- Replay validation
- Replay reporting
- Replay traceability
- Replay audit support
- Replay governance controls

The Replay Module operates across the complete TODP lifecycle:

Opportunity

→ Authority

→ Evidence

→ Retrieval

→ Ranking

→ Recommendation

→ Governance

→ Audit

→ Replay

4. Architectural Decision ADR-0112

Decision Title

Replay as Independent Domain Module

Status

Approved

Decision

The Replay Module shall operate as an independent architectural domain with dedicated ownership, governance controls, operational oversight, and lifecycle management.

Rationale

Replay capabilities require specialized controls and operate across all platform domains.

Consequences

Positive

- Independent evolution
- Clear accountability
- Reduced architectural coupling

Negative

- Additional integration complexity

5. Architectural Decision ADR-0113

Decision Title

Replay as Core Platform Capability

Status

Approved

Decision

Replay shall be considered a mandatory core platform capability rather than an optional feature.

Rationale

Replayability is a foundational TODP principle supporting transparency, auditability, explainability, and governance.

Consequences

Positive

- Strong accountability
- Enhanced trust

Negative

- Increased data preservation requirements

6. Architectural Decision ADR-0114

Decision Title

Full Lifecycle Replay Architecture

Status

Approved

Decision

Replay operations shall support reconstruction of the complete TODP workflow.

Replay Coverage

- Opportunity processing
- Authority evaluation
- Evidence lifecycle
- Retrieval operations
- Ranking evaluations
- Recommendation generation
- Governance reviews
- Audit activities

Rationale

Supports complete decision transparency.

Consequences

Positive

- End-to-end visibility

Negative

- Increased retention complexity

7. Architectural Decision ADR-0115

Decision Title

Historical State Preservation Requirement

Status

Approved

Decision

All replayable entities shall preserve historical states required for reconstruction.

Preserved Artifacts

- Opportunity states
- Authority states
- Evidence versions
- Retrieval criteria
- Ranking configurations
- Recommendation versions
- Governance rules
- Audit records

Rationale

Supports accurate replay execution.

Consequences

Positive

- Improved replay accuracy

Negative

- Long-term storage requirements

8. Architectural Decision ADR-0116

Decision Title

Version-Aware Replay Execution

Status

Approved

Decision

Replay operations shall execute using the historical versions that existed at the time of the original decision.

Version Categories

- Policy versions
- Rule versions
- Ranking versions
- Recommendation versions
- Governance versions

Rationale

Ensures accurate historical reconstruction.

Consequences

Positive

- Replay fidelity

Negative

- Additional version management

9. Architectural Decision ADR-0117

Decision Title

Replay Traceability Framework

Status

Approved

Decision

Replay executions shall maintain complete traceability.

Required Traceability Elements

- Replay identifier
- Opportunity identifier
- Historical references
- Version references
- Execution timestamps
- Replay requester

Rationale

Supports auditability and accountability.

Consequences

Positive

- Complete replay visibility

Negative

- Additional metadata management

10. Architectural Decision ADR-0118

Decision Title

Governance-Controlled Replay Access

Status

Approved

Decision

Replay operations shall be subject to governance controls and authorization validation.

Access Requirements

- Authority validation
- Governance approval where applicable
- Audit logging
- Purpose recording

Rationale

Protects sensitive historical information.

Consequences

Positive

- Improved governance

Negative

- Additional approval workflows

11. Architectural Decision ADR-0119

Decision Title

Replay Outcome Comparison Capability

Status

Approved

Decision

The Replay Module shall support comparison between historical outcomes and replay outcomes.

Comparison Areas

- Ranking outcomes
- Recommendation outcomes

- Governance decisions
- Evidence availability

Rationale

Supports investigations and validation.

Consequences

Positive

- Better analytical capabilities

Negative

- Additional processing requirements

12. Architectural Decision ADR-0120

Decision Title

Replay Validation Framework

Status

Approved

Decision

Replay executions shall include validation procedures.

Validation Categories

- Data integrity validation
- Version integrity validation
- Governance validation
- Traceability validation

Rationale

Improves replay reliability.

Consequences

Positive

- Greater replay confidence

Negative

- Additional execution overhead

13. Architectural Decision ADR-0121

Decision Title

Replay Result Preservation

Status

Approved

Decision

Replay outputs shall be preserved as immutable records.

Preserved Elements

- Replay request
- Replay configuration
- Replay outcome
- Validation results
- Comparison results

Rationale

Supports audits and investigations.

Consequences

Positive

- Historical replay visibility

Negative

- Increased retention volume

14. Architectural Decision ADR-0122

Decision Title

Replay Investigation Support

Status

Approved

Decision

The Replay Module shall support forensic and investigative activities.

Capabilities

- Timeline reconstruction
- Decision reconstruction
- Governance validation
- Audit support
- Historical analysis

Rationale

Supports enterprise oversight requirements.

Consequences

Positive

- Improved investigative capability

Negative

- Additional operational complexity

15. Architectural Decision ADR-0123

Decision Title

Replay Reporting Framework

Status

Approved

Decision

Replay operations shall generate reporting artifacts.

Report Categories

- Replay execution reports
- Validation reports
- Investigation reports
- Governance reports
- Compliance reports

Rationale

Supports transparency and accountability.

Consequences

Positive

- Better visibility

Negative

- Reporting management effort

16. Architectural Decision ADR-0124

Decision Title

Replay Observability Framework

Status

Approved

Decision

Replay operations shall generate telemetry supporting operational monitoring.

Required Telemetry

- Replay volume
- Replay latency
- Validation failures
- Comparison outcomes
- Resource utilization

Rationale

Supports operational management.

Consequences

Positive

- Improved monitoring

Negative

- Additional telemetry storage

17. Architectural Decision ADR-0125

Decision Title

Scalable Replay Architecture

Status

Approved

Decision

Replay architecture shall support increasing historical volumes and investigation workloads without requiring fundamental redesign.

Scalability Targets

- Growing historical records
- Growing audit requests
- Growing investigations
- Enterprise-scale adoption

Rationale

Supports long-term platform growth.

Consequences

Positive

- Enterprise scalability

Negative

- Capacity planning requirements

18. Architectural Constraints

The Replay Module shall comply with:

RPL-CON-001

Support full lifecycle reconstruction.

RPL-CON-002

Maintain historical fidelity.

RPL-CON-003

Preserve version awareness.

RPL-CON-004

Support governance-controlled access.

RPL-CON-005

Maintain replay traceability.

RPL-CON-006

Preserve audit visibility.

RPL-CON-007

Protect replay integrity.

RPL-CON-008

Support enterprise scalability.

19. Metrics

The following metrics shall be monitored:

MET-RPL-001

Replay execution volume.

MET-RPL-002

Replay latency.

MET-RPL-003

Replay success rate.

MET-RPL-004

Replay validation pass rate.

MET-RPL-005

Replay comparison accuracy.

MET-RPL-006

Replay traceability completeness.

MET-RPL-007

Historical reconstruction accuracy.

Replay governance compliance rate.

20. Audit Requirements

Audits shall verify:

- Replay completeness
- Historical fidelity
- Version preservation
- Replay access controls
- Validation compliance
- Replay traceability
- Investigation readiness
- Reporting completeness

Audit evidence shall be retained according to enterprise retention policies.

21. Compliance Requirements

The Replay Module architecture shall support:

- SOC 2 accountability controls
- SOC 2 auditability requirements
- ISO 27001 logging and monitoring controls
- ISO 27001 investigation support requirements
- Enterprise audit programs
- Customer assurance reviews
- Regulatory review activities where applicable

Compliance mappings shall be maintained through RTM, SEC, GOV, AUD, COMP, and REL artifacts.

22. Section Summary

This section defines the approved architecture decisions governing the Replay Module, including full lifecycle reconstruction, historical state preservation, version-aware replay execution, governance-controlled access, replay validation, comparison capabilities, investigation support, replay reporting, observability requirements, audit obligations, and compliance requirements. These decisions establish Replay as the transparency and reconstruction layer of TODP and ensure that historical decision outcomes remain explainable, traceable, auditable, accountable, and reproducible throughout the platform lifecycle.

ADR V1 – Architecture Decision Record Repository

Section 13: Cross-Cutting Architecture Decisions

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Document Title: Cross-Cutting Architecture Decisions

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Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

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1. Purpose

The purpose of this section is to document architecture decisions that apply across all TODP modules and platform capabilities.

These decisions establish common architectural standards, enterprise-wide controls, shared design principles, interoperability requirements, observability expectations, governance obligations, security controls, compliance requirements, and operational standards that transcend individual module boundaries.

The decisions defined within this section are mandatory for:

- Opportunity
- Authority
- Evidence
- Retrieval
- Ranking
- Recommendation
- Governance
- Audit
- Replay

2. Objectives

OBJ-XARCH-001

Establish platform-wide architectural consistency.

OBJ-XARCH-002

Reduce architectural fragmentation.

OBJ-XARCH-003

Provide enterprise-grade governance controls.

OBJ-XARCH-004

Support interoperability between modules.

OBJ-XARCH-005

Maintain platform-wide traceability.

OBJ-XARCH-006

Ensure auditability across the platform.

OBJ-XARCH-007

Support future scalability.

OBJ-XARCH-008

Preserve transparency and replayability.

3. Scope

This section governs:

- Platform-wide architecture standards
- Cross-module communication
- Shared services
- Identity and access integration
- Data governance integration
- Observability standards
- Audit standards
- Security standards
- Compliance standards
- Lifecycle management standards

The decisions apply to all current and future TODP platform components.

4. Architectural Decision ADR-0126

Decision Title

Modular Domain-Oriented Architecture

Status

Approved

Decision

TODP shall maintain a modular domain-oriented architecture where each core module remains independently governed and independently evolvable.

Governed Domains

- Opportunity
- Authority
- Evidence
- Retrieval
- Ranking
- Recommendation
- Governance
- Audit
- Replay

Rationale

Supports maintainability, scalability, and clear ownership.

Consequences

Positive

- Reduced coupling
- Better scalability
- Independent evolution

Negative

- Additional interface governance

5. Architectural Decision ADR-0127

Decision Title

API-Driven Interoperability Model

Status

Approved

Decision

Cross-module communication shall occur through approved and governed APIs.

Requirements

- Versioned interfaces
- Defined contracts
- Controlled dependencies
- Traceable interactions

Rationale

Provides interoperability while preserving modularity.

Consequences

Positive

- Strong integration governance

Negative

- API lifecycle management effort

6. Architectural Decision ADR-0128

Decision Title

Enterprise Traceability Framework

Status

Approved

Decision

All platform activities shall support end-to-end traceability.

Traceability Anchors

- Opportunity ID
- Authority ID
- Evidence ID
- Retrieval ID
- Ranking ID
- Recommendation ID
- Governance ID
- Audit ID
- Replay ID

Rationale

Supports transparency and investigations.

Consequences

Positive

- Complete lifecycle visibility

Negative

- Metadata overhead

7. Architectural Decision ADR-0129

Decision Title

Universal Explainability Requirement

Status

Approved

Decision

All platform decisions and evaluations shall provide explainable outputs.

Applicable Areas

- Retrieval outcomes
- Ranking outcomes
- Recommendations
- Governance decisions
- Replay outcomes

Rationale

Supports trust, transparency, and accountability.

Consequences

Positive

- Improved stakeholder confidence

Negative

- Additional processing complexity

8. Architectural Decision ADR-0130

Decision Title

Centralized Identity and Access Governance

Status

Approved

Decision

Identity and authorization controls shall be centralized through Authority governance mechanisms.

Rationale

Provides consistent access control enforcement.

Consequences

Positive

- Simplified governance
- Reduced security risk

Negative

- Centralized dependency

9. Architectural Decision ADR-0131

Decision Title

Immutable Auditability Standard

Status

Approved

Decision

Material platform activities shall generate immutable audit records.

Covered Activities

- Administrative actions
- Business actions
- Governance actions
- Security actions
- Replay actions

Rationale

Supports compliance and investigations.

Consequences

Positive

- Strong audit integrity

Negative

- Increased retention requirements

10. Architectural Decision ADR-0132

Decision Title

Replayability by Design

Status

Approved

Decision

Replay requirements shall be considered during architecture design activities.

Design Requirements

- State preservation
- Version preservation
- Event preservation
- Traceability preservation

Rationale

Replay is a foundational TODP capability.

Consequences

Positive

- Reliable reconstruction

Negative

- Increased architecture complexity

11. Architectural Decision ADR-0133

Decision Title

Version-Controlled Configuration Management

Status

Approved

Decision

Material platform configurations shall be version controlled.

Configuration Categories

- Governance rules
- Ranking configurations
- Recommendation configurations
- Security configurations
- Operational configurations

Rationale

Supports auditability and change control.

Consequences

Positive

- Controlled evolution

Negative

- Additional administration effort

12. Architectural Decision ADR-0134

Decision Title

Platform-Wide Observability Framework

Status

Approved

Decision

All modules shall produce standardized telemetry.

Required Telemetry Types

- Metrics
- Logs
- Events
- Traces
- Health indicators

Rationale

Supports operational management.

Consequences

Positive

- Improved monitoring

Negative

- Additional storage requirements

13. Architectural Decision ADR-0135

Decision Title

Security-by-Design Architecture Standard

Status

Approved

Decision

Security requirements shall be incorporated into all architecture decisions.

Security Areas

- Authentication
- Authorization
- Encryption
- Logging
- Monitoring
- Incident response

Rationale

Supports enterprise security objectives.

Consequences

Positive

- Reduced security risk

Negative

- Additional design effort

14. Architectural Decision ADR-0136

Decision Title

Compliance-by-Design Architecture Standard

Status

Approved

Decision

Compliance requirements shall be evaluated during architecture decision-making.

Applicable Frameworks

- SOC 2
- ISO 27001
- Internal Governance Standards
- Customer Security Requirements

Rationale

Reduces future compliance remediation effort.

Consequences

Positive

- Improved compliance readiness

Negative

- Additional review activities

15. Architectural Decision ADR-0137

Decision Title

Event-Centric Activity Recording

Status

Approved

Decision

Material platform activities shall generate traceable events.

Required Event Attributes

- Identifier
- Timestamp
- Actor

- Action
- Context
- Outcome

Rationale

Supports auditability and replayability.

Consequences

Positive

- Strong event lineage

Negative

- Increased event volume

16. Architectural Decision ADR-0138

Decision Title

Data Ownership and Stewardship Model

Status

Approved

Decision

All major platform entities shall maintain assigned ownership and stewardship responsibilities.

Applicable Entities

- Opportunities
- Evidence
- Governance records
- Audit records
- Configuration artifacts

Rationale

Supports accountability and operational governance.

Consequences

Positive

- Improved ownership clarity

Negative

- Governance administration overhead

17. Architectural Decision ADR-0139

Decision Title

Enterprise Scalability Standard

Status

Approved

Decision

Architecture decisions shall consider future enterprise-scale growth.

Growth Areas

- Users
- Opportunities
- Evidence
- Retrieval requests
- Recommendations
- Governance reviews
- Audit events

Rationale

Supports long-term business objectives.

Consequences

Positive

- Reduced future redesign

Negative

- Additional capacity planning

18. Architectural Decision ADR-0140

Decision Title

Operational Resilience Standard

Status

Approved

Decision

Architecture decisions shall incorporate resilience considerations.

Resilience Areas

- Fault tolerance
- Recovery
- Availability
- Backup
- Disaster recovery

Rationale

Supports operational continuity.

Consequences

Positive

- Improved reliability

Negative

- Additional infrastructure requirements

19. Architectural Decision ADR-0141

Decision Title

Enterprise Change Governance Framework

Status

Approved

Decision

Material architecture changes shall require governance review and approved ADRs.

Governed Changes

- Architecture modifications
- Policy changes
- Security changes
- Data model changes
- Integration changes

Rationale

Maintains architecture consistency.

Consequences

Positive

- Controlled change management

Negative

- Additional governance processes

20. Architectural Constraints

The platform shall comply with:

XARCH-CON-001

Maintain modular architecture.

XARCH-CON-002

Support API-driven integration.

XARCH-CON-003

Preserve traceability.

XARCH-CON-004

Maintain explainability.

XARCH-CON-005

Support auditability.

XARCH-CON-006

Support replayability.

XARCH-CON-007

Enforce governance controls.

XARCH-CON-008

Support enterprise scalability.

21. Metrics

The following metrics shall be monitored:

MET-XARCH-001

Cross-module integration success rate.

MET-XARCH-002

Traceability completeness.

MET-XARCH-003

Audit coverage.

MET-XARCH-004

Replay coverage.

MET-XARCH-005

Configuration governance compliance.

MET-XARCH-006

Platform observability coverage.

MET-XARCH-007

Architecture exception rate.

MET-XARCH-008

Scalability readiness score.

22. Audit Requirements

Audits shall verify:

- Architecture standard compliance
- Integration governance compliance
- Traceability completeness
- Audit coverage
- Replay readiness
- Security compliance
- Configuration governance
- Observability coverage

Audit findings shall be formally documented and remediated.

23. Compliance Requirements

The cross-cutting architecture standards shall support:

- SOC 2
- ISO 27001
- ISO 9001
- Enterprise Governance Policies
- Customer Security Reviews
- Procurement Assessments
- Internal Audit Programs

Compliance mappings shall be maintained across RTM, SEC, GOV, AUD, REL, and COMP artifacts.

24. Section Summary

This section defines the cross-cutting architecture decisions governing all TODP modules, including modular architecture standards, API-driven interoperability, traceability requirements, explainability standards, auditability controls, replayability requirements, security-by-design principles, compliance-by-design principles, observability standards, operational resilience requirements, scalability expectations, and enterprise change governance controls. These decisions establish a consistent architectural foundation across the entire TODP platform and ensure long-term enterprise readiness, transparency, governance, auditability, and operational sustainability.

ADR V1 – Architecture Decision Record Repository

Section 14: Architecture Risk & Exception Management

Document ID: ADR-V1-S14

Document Title: Architecture Risk & Exception Management

Version: 1.0

Status: Approved

Owner: Enterprise Architecture Function

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to define the architecture risk management framework, exception governance model, risk assessment methodology, risk ownership requirements, mitigation controls, escalation procedures, and audit obligations applicable to the Transparent Opportunity Decision Platform (TODP).

This framework ensures that architectural risks are identified, evaluated, documented, mitigated, monitored, governed, and periodically reviewed throughout the platform lifecycle.

The framework also establishes formal processes for managing architecture exceptions where temporary deviations from approved architecture standards become necessary.

2. Objectives

OBJ-RISK-001

Provide structured architecture risk governance.

OBJ-RISK-002

Identify and manage architecture risks proactively.

OBJ-RISK-003

Support risk-informed decision making.

OBJ-RISK-004

Control architecture exceptions.

OBJ-RISK-005

Maintain architecture compliance.

OBJ-RISK-006

Support audit and regulatory reviews.

OBJ-RISK-007

Provide accountability for architecture risks.

OBJ-RISK-008

Enable continuous risk monitoring.

3. Scope

This section applies to:

- Architecture decisions
- Technology selections
- Infrastructure architecture
- Data architecture
- Security architecture
- Integration architecture
- AI architecture
- Governance architecture

- Operational architecture
- Compliance architecture

The framework applies across:

- Opportunity
- Authority
- Evidence
- Retrieval
- Ranking
- Recommendation
- Governance
- Audit
- Replay

4. Architecture Risk Governance Model

Architecture risks shall be managed through a formal governance structure.

Governance Bodies

Architecture Review Board (ARB)

Primary architecture governance authority.

Product Governance Council

Executive governance authority.

Security Governance Team

Security risk authority.

Compliance Governance Team

Compliance risk authority.

Engineering Leadership

Implementation risk authority.

Risk ownership shall be assigned for every identified architecture risk.

5. Architecture Risk Classification Framework

Architecture risks shall be classified using the following categories:

AR-RISK-001

Strategic Architecture Risk

AR-RISK-002

Technology Risk

AR-RISK-003

Security Risk

AR-RISK-004

Compliance Risk

AR-RISK-005

Operational Risk

AR-RISK-006

Scalability Risk

AR-RISK-007

Availability Risk

AR-RISK-008

Data Risk

AR-RISK-009

Integration Risk

AR-RISK-010

Governance Risk

Each risk shall be assigned one primary category.

6. Risk Assessment Methodology

All architecture risks shall be assessed using standardized criteria.

Assessment Factors

- Likelihood
- Impact
- Detectability
- Duration
- Recoverability
- Business effect

- Customer effect
- Compliance effect

Risk assessments shall be documented and reviewed by the Architecture Review Board.

7. Risk Severity Model

The following severity model shall be used:

Critical

Immediate enterprise impact.

High

Significant business impact.

Medium

Manageable operational impact.

Low

Limited impact.

Informational

Monitoring only.

Risk severity assignments shall be periodically reassessed.

8. Risk Ownership Requirements

Every architecture risk shall have assigned ownership.

Required Roles

Risk Owner

Accountable party.

Risk Reviewer

Independent reviewer.

Mitigation Owner

Responsible for remediation.

Governance Approver

Approves risk acceptance.

Ownership assignments shall be maintained within the architecture risk register.

9. Risk Lifecycle Management

Architecture risks shall follow the following lifecycle:

Identified

Risk discovered.

Assessed

Risk evaluated.

Approved

Risk accepted for tracking.

Mitigated

Controls implemented.

Monitored

Ongoing monitoring.

Closed

Risk retired.

Archived

Historical retention.

All lifecycle transitions shall be auditable.

10. Risk Mitigation Requirements

Risk mitigation plans shall include:

- Risk description
- Root cause
- Mitigation actions
- Target completion date
- Responsible owner
- Success criteria
- Residual risk assessment

Mitigation effectiveness shall be periodically reviewed.

11. Residual Risk Management

Residual risks remaining after mitigation shall be formally documented.

Requirements

- Residual risk assessment
- Business justification
- Governance approval
- Monitoring plan

Residual risks shall not be assumed without approval.

12. Exception Management Framework

Architecture exceptions shall be managed through a formal exception process.

Exceptions may only be granted when:

- Business necessity exists
- Alternative options were evaluated
- Risk assessment completed
- Governance approval obtained

Unauthorized exceptions are prohibited.

13. Exception Classification Model

Exception categories shall include:

ARC-EXC-001

Technology Exception

ARC-EXC-002

Security Exception

ARC-EXC-003

Compliance Exception

ARC-EXC-004

Operational Exception

ARC-EXC-005

Governance Exception

ARC-EXC-006

Integration Exception

ARC-EXC-007

Data Exception

ARC-EXC-008

Infrastructure Exception

Each exception shall be documented and tracked.

14. Exception Approval Requirements

Exception approvals shall require:

Architecture Review Board

Required

Security Review

Required for security-impacting exceptions

Compliance Review

Required for compliance-impacting exceptions

Product Governance Council

Required for high-risk exceptions

Approval evidence shall be retained for audit purposes.

15. Exception Lifecycle Management

Architecture exceptions shall follow the lifecycle:

Requested

Under Review

Approved

Implemented

Monitored

Expired

Closed

Renewed

Exception expiration dates are mandatory.

16. Exception Documentation Requirements

Every exception record shall include:

- Exception identifier
- Requestor
- Business justification
- Risk assessment
- Alternative analysis
- Approval records
- Expiration date
- Monitoring plan

Incomplete exception requests shall not be approved.

17. Risk Monitoring Framework

Architecture risks shall be continuously monitored.

Monitoring Categories

- Security exposure
- Compliance exposure
- Performance degradation
- Scalability limitations
- Availability concerns
- Governance failures

Monitoring results shall be reported to governance stakeholders.

18. Escalation Framework

The following escalation paths shall be maintained:

Low Risk

Architecture Team

Medium Risk

Architecture Review Board

High Risk

Product Governance Council

Critical Risk

Executive Leadership

Escalations shall be documented and tracked.

19. Risk Reporting Requirements

Risk reporting shall include:

- Open risks
- Mitigated risks
- Residual risks
- Exception inventory
- Exception expirations
- Emerging risks
- Risk trends

Reports shall be generated on a recurring basis.

20. Architecture Constraints

Risk and exception management shall comply with:

RISK-CON-001

Maintain risk ownership.

RISK-CON-002

Support auditability.

RISK-CON-003

Require governance approval.

RISK-CON-004

Maintain exception traceability.

RISK-CON-005

Support compliance requirements.

RISK-CON-006

Preserve historical records.

RISK-CON-007

Enforce lifecycle management.

RISK-CON-008

Support enterprise oversight.

21. Metrics

The following metrics shall be monitored:

MET-RISK-001

Open architecture risks.

MET-RISK-002

Critical risk count.

MET-RISK-003

Mitigation completion rate.

MET-RISK-004

Exception volume.

MET-RISK-005

Expired exception count.

MET-RISK-006

Residual risk volume.

MET-RISK-007

Risk reassessment completion rate.

MET-RISK-008

Governance review compliance rate.

22. Audit Requirements

Audits shall verify:

- Risk identification completeness
- Risk ownership assignments
- Mitigation effectiveness
- Residual risk approvals
- Exception governance compliance
- Exception expiration management
- Escalation procedures
- Reporting completeness

Audit findings shall be documented and tracked to closure.

23. Compliance Requirements

The architecture risk and exception management framework shall support:

- SOC 2 risk management requirements
- ISO 27001 risk assessment requirements
- ISO 27001 risk treatment controls
- Enterprise governance policies
- Internal audit programs
- Customer due diligence reviews
- Regulatory assessment activities where applicable

Compliance mappings shall be maintained through RTM, SEC, GOV, AUD, COMP, and REL artifacts.

24. Section Summary

This section defines the architecture risk and exception management framework governing TODP, including risk classification, risk assessment methodology, ownership requirements, mitigation planning, residual risk management, exception governance, approval processes, lifecycle management, monitoring controls, escalation procedures, reporting requirements, audit obligations, and compliance requirements. These controls ensure that architecture risks and deviations are managed transparently, governed consistently, audited effectively, and aligned with enterprise risk tolerance throughout the platform lifecycle.

ADR V1 – Architecture Decision Record Repository

Section 15: ADR Governance Summary & Repository Maintenance

Document ID: ADR-V1-S15

Document Title: ADR Governance Summary & Repository Maintenance

Version: 1.0

Status: Approved

Owner: Enterprise Architecture Function

Review Authority: Architecture Review Board (ARB)

Approval Authority: Product Governance Council

Classification: Internal Confidential

1. Purpose

The purpose of this section is to establish the long-term governance, operational maintenance, review procedures, repository sustainability requirements, ownership model, audit controls, and continuous improvement framework for the Architecture Decision Record (ADR) repository within the Transparent Opportunity Decision Platform (TODP).

This section serves as the governing framework ensuring that the ADR repository remains complete, accurate, traceable, auditable, current, and aligned with evolving business, technical, security, governance, operational, and compliance requirements.

2. Objectives

OBJ-ADR-GOV-001

Maintain ADR repository integrity.

OBJ-ADR-GOV-002

Ensure ongoing architecture governance.

OBJ-ADR-GOV-003

Preserve architectural knowledge.

OBJ-ADR-GOV-004

Support enterprise audit readiness.

OBJ-ADR-GOV-005

Enable continuous architecture improvement.

OBJ-ADR-GOV-006

Ensure repository sustainability.

OBJ-ADR-GOV-007

Support future platform evolution.

OBJ-ADR-GOV-008

Maintain enterprise architecture accountability.

3. Scope

This section governs:

- ADR repository operations
- Repository ownership
- Repository maintenance
- ADR lifecycle governance
- Architecture review activities
- Repository audits
- Repository reporting
- Repository retention
- Continuous improvement activities

The scope applies to all ADRs governing:

- Opportunity
- Authority
- Evidence
- Retrieval
- Ranking
- Recommendation
- Governance
- Audit
- Replay

4. ADR Governance Authority Model

The ADR repository shall be governed through the following authority structure.

Enterprise Architecture Function

Repository Owner

Architecture Review Board (ARB)

Architecture Governance Authority

Product Governance Council

Executive Approval Authority

Security Governance Team

Security Review Authority

Compliance Governance Team

Compliance Review Authority

Engineering Leadership

Implementation Authority

The authority model shall be reviewed annually.

5. Repository Ownership Responsibilities

Repository owners shall be responsible for:

- Repository integrity
- Repository completeness
- ADR quality standards
- Lifecycle governance
- Audit support
- Traceability maintenance
- Repository reporting
- Continuous improvement

Ownership responsibilities shall be formally documented.

6. ADR Review Framework

All approved ADRs shall undergo periodic review.

Review Triggers

- Annual review cycle
- Major architecture changes
- Security incidents
- Compliance changes
- Technology replacement
- Organizational changes

Review outcomes shall be documented.

7. ADR Change Governance

Changes to approved ADRs shall require governance review.

Governed Changes

- Decision modifications
- Decision retirement
- Decision supersession
- Decision deprecation
- Repository structure changes

Unapproved ADR modifications are prohibited.

8. ADR Supersession Management

Architecture decisions may be superseded through approved governance processes.

Supersession Requirements

- New ADR reference
- Supersession rationale
- Impact assessment
- Approval evidence
- Effective date

Historical ADRs shall remain accessible.

9. ADR Deprecation Framework

Deprecated ADRs shall remain retained for historical and audit purposes.

Deprecation Conditions

- Technology retirement
- Architecture replacement
- Business capability retirement
- Governance changes

Deprecation records shall maintain complete historical context.

10. Repository Maintenance Procedures

Repository maintenance activities shall include:

- Metadata validation
- Traceability validation
- Version verification
- Approval verification
- Integrity validation
- Archive management

Maintenance procedures shall be performed periodically.

11. Repository Quality Assurance

The ADR repository shall maintain quality controls.

Quality Dimensions

- Completeness
- Accuracy
- Consistency
- Traceability
- Auditability
- Governance alignment

Quality reviews shall be documented.

12. Repository Traceability Maintenance

Repository traceability shall be continuously maintained.

Required Relationships

ADR → BRD

ADR → SRS

ADR → SAD

ADR → TDD

ADR → TEST

ADR → RTM

ADR → SEC

ADR → GOV

ADR → AUD

ADR → REL

Broken traceability relationships shall be remediated.

13. Repository Integrity Controls

The repository shall implement integrity protections.

Controls

- Version control
- Change history preservation
- Access controls
- Approval records
- Audit logging

Repository integrity violations shall be investigated.

14. Repository Retention Governance

ADR repository records shall be retained according to approved retention policies.

Retention Categories

- Active ADRs
- Superseded ADRs
- Deprecated ADRs
- Archived ADRs
- Approval records
- Audit records

Architecture decision history shall be preserved permanently unless legal obligations require otherwise.

15. Architecture Knowledge Management

The ADR repository shall serve as the authoritative architecture knowledge base.

Knowledge Areas

- Decision rationale
- Tradeoff analysis
- Architecture evolution
- Governance history
- Risk decisions
- Technology decisions

Knowledge preservation shall be treated as a strategic objective.

16. Repository Audit Framework

The ADR repository shall support recurring audits.

Audit Areas

- ADR completeness
- Governance compliance
- Traceability coverage
- Approval compliance
- Retention compliance
- Repository integrity

Audit findings shall be tracked through remediation workflows.

17. Repository Reporting Framework

Repository reporting shall include:

- ADR inventory
- ADR lifecycle status
- Superseded ADRs
- Open reviews
- Traceability coverage

- Governance compliance
- Audit findings

Reports shall be generated on a recurring basis.

18. Continuous Improvement Framework

The ADR governance process shall support continuous improvement.

Improvement Sources

- Audit findings
- Security reviews
- Compliance reviews
- Operational reviews
- Architecture retrospectives
- Customer feedback

Improvement actions shall be tracked to completion.

19. Architecture Governance Maturity Management

Architecture governance maturity shall be periodically assessed.

Assessment Areas

- Documentation quality
- Governance effectiveness
- Traceability maturity
- Audit readiness
- Compliance readiness
- Operational sustainability

Assessment results shall inform improvement initiatives.

20. Repository Sustainability Requirements

The repository shall support long-term sustainability.

Requirements

- Knowledge preservation
- Personnel transitions
- Technology evolution
- Governance continuity
- Historical accessibility

Repository sustainability shall be incorporated into strategic planning.

21. Architectural Constraints

ADR governance and repository maintenance shall comply with:

ADR-CON-001

Maintain repository integrity.

ADR-CON-002

Preserve architecture history.

ADR-CON-003

Maintain traceability.

ADR-CON-004

Support auditability.

ADR-CON-005

Enforce governance controls.

ADR-CON-006

Support compliance obligations.

ADR-CON-007

Preserve institutional knowledge.

ADR-CON-008

Support long-term sustainability.

22. Metrics

The following metrics shall be monitored:

MET-ADR-001

ADR review completion rate.

MET-ADR-002

Traceability completeness rate.

MET-ADR-003

Repository quality score.

MET-ADR-004

Repository audit findings.

MET-ADR-005

Governance compliance rate.

MET-ADR-006

ADR supersession volume.

MET-ADR-007

Repository integrity incidents.

MET-ADR-008

Continuous improvement completion rate.

23. Audit Requirements

Audits shall verify:

- Repository completeness
- Review compliance
- Traceability maintenance
- Governance effectiveness
- Integrity controls
- Retention compliance
- Knowledge preservation
- Continuous improvement execution

Audit findings shall be documented and remediated through approved governance processes.

24. Compliance Requirements

ADR governance and repository maintenance shall support:

- SOC 2 governance requirements
- ISO 27001 documentation controls
- ISO 27001 change management controls
- ISO 9001 quality management requirements
- Enterprise governance standards
- Internal audit programs
- Customer assurance reviews
- Regulatory assessment activities where applicable

Compliance mappings shall be maintained through RTM, SEC, GOV, AUD, COMP, and REL artifacts.

25. ADR V1 Summary

ADR V1 establishes the authoritative architectural governance framework for the Transparent Opportunity Decision Platform. The document defines architecture decision governance, repository standards, decision-making principles, module-specific architecture decisions, cross-cutting architecture standards, risk and exception management controls, and long-term repository maintenance requirements.

The approved architecture establishes the following governed platform flow:

Opportunity

- Authority
- Evidence
- Retrieval
- Ranking
- Recommendation
- Governance
- Audit
- Replay

The architecture is governed by the core principles of:

- Transparency
- Explainability
- Traceability
- Governance
- Auditability
- Replayability
- Enterprise Readiness

ADR V1 serves as the authoritative architectural decision record repository and shall remain the primary source of truth for architecture governance, architecture evolution, architecture compliance, and architecture audit activities throughout the lifecycle of TODP.